RESIDUAL-DIMENSION OBSTRUCTIONS FOR NULL-PROJECTED SUPERSYMMETRIC YANG–MILLS THEORIES

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Abstract. *We classify null-projected supersymmetric Yang–Mills theories with a Majorana–Weyl gaugino in signature (s,t), defined by five conditions on the gaugino constraint, the transformation rules, the residual gauge field, the translation bracket and the absorption of the cubic gaugino terms. Such a theory built on k null vectors exists if and only if $s+t-2k=10$ and $s-t\in\{8,0,-8\}$, or $s+t-2k=2$ and $s=t$. The key step is a converse of the cubic gamma-matrix identity: symmetric gamma matrices that satisfy it act on spinors of dimension $2(d-2)$. Hence the signatures $(9+8n+q)+(1+q)$ with $n\geq 1$ and $0\leq q\leq 3$ proposed by Rios, Marrani and Chester, among them $27+3$, carry no such theory, and neither another contraction of the source in the gauge field equation nor several gauginos with their own parameters change this. In the single-time signatures $(9+8n)+1$ no unitary massless multiplet without tensor charges consists of scalars, spinors and vectors.*

1. Introduction

In more than two dimensions, conventional minimal supersymmetric Yang–Mills theory with only a gauge field and a gaugino exists for a single time in three, four, six and ten dimensions [1, 2, 3, 4]. The reason is one algebraic fact. Varying the gauge field inside the covariant derivative of the gaugino kinetic term produces a term cubic in the gaugino, and no other term in the variation of the Lagrangian can cancel it. This term vanishes precisely when the symmetric gamma matrices obey the Fierz identity

$$\gamma^{\mu}_{\alpha\beta}\gamma_{\mu\,\gamma\delta}+\gamma^{\mu}_{\alpha\gamma}\gamma_{\mu\,\beta\delta}+\gamma^{\mu}_{\alpha\delta}\gamma_{\mu\,\beta\gamma}=0, \tag{1.1}$$

which holds for the Majorana spinors of three and four dimensions, the Weyl spinors of six and the Majorana–Weyl spinors of ten, the spinors attached to the four normed division algebras [2, 3, 4]. The same identity makes the Green–Schwarz superstring κ-symmetric [5], its analogues for forms of higher degree produce the scan of supersymmetric branes [6, 7], and Blencowe and Duff extended that scan to spacetimes with several times [8]. Above ten dimensions the spinors are too large: a Majorana–Weyl spinor in D dimensions has $2^{D/2-1}$ components, while (1.1) allows $2(D-2)$ of them (Theorem 3.3).

The null-frame constructions considered here extend this field content by introducing additional timelike as well as spacelike directions. Signatures with several times have been studied for several reasons. F-theory points to a twelve-dimensional origin of type IIB string theory [9]; Bars proposed an S-theory in $11+2$ dimensions and made a case for fourteen [10, 11]; in two-time physics an $Sp(2,\mathbb{R})$ gauge symmetry removes one time and one space direction [12, 13, 14]; and the dualities of string theory and M-theory relate theories of different signatures [15, 16]. Superparticles beyond eleven dimensions were studied by Rudychev and Sezgin [17], and supergravity in $10+2$ dimensions by Nishino [18]. Supersymmetric Yang–Mills equations in $10+2$ dimensions were written by Nishino and Sezgin [19], a theory in $11+3$ dimensions was given by Sezgin [20], and the mechanism behind these constructions was set out by Rudychev, Sezgin and Sundell [21]. Nishino then

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built supersymmetric Yang–Mills theories in every signature $(9+m)+(1+m)$ [22]. His construction introduces m mutually orthogonal null vectors $n_{(1)},\dots,n_{(m)}$, constrains the gaugino by $\not{n}_{(i)}\lambda=0$, and lets the supersymmetry algebra close on translations contracted with $n_{(1)}\wedge\cdots\wedge n_{(m)}$. The cubic gaugino terms then need not vanish. They only have to take a form that can be absorbed into the field equation of the gauge field, and in Nishino's construction they do.

Rios, Marrani and Chester used the periodic reality and symmetry properties of spinors to propose that Nishino's existence proofs could be transferred to new signatures [23, Sec. III.A]. Their proposal keeps $D=s+t$ and $s-t$ modulo eight fixed while moving $4n$ timelike directions into space. The resulting four families are

$$(9+8n)+1,\qquad (10+8n)+2,\qquad (11+8n)+3,\qquad (12+8n)+4, \tag{1.2}$$

with the chiral superalgebras of their eqs. (3.5)–(3.8). They called these theories exceptional super Yang–Mills theories. The signature $27+3$, the case $n=2$ of the third family, was then proposed as the home of a theory whose fields fill the exceptional Jordan algebra and whose worldvolume would carry M-theory [24]. The same theory was used to model dark matter [25], and it entered the construction of "monstrous" M-theory [26] and the programme of exceptional periodicity [27]. Here we test this proposed transfer within the explicit transformation, projection and absorption assumptions (N1)–(N5). Proposition 7.1 writes out what repeating Nishino's proof in $(11+8n)+3$ requires and shows that each choice of the null frame fails: Nishino's rank does not fit, the translation bracket allows only rank two, and at rank two the cubic terms cannot be absorbed.

The transfer does not work under (N1)–(N5). Reality and symmetry of spinors are not the only data that enter this mechanism. The null vectors must be mutually orthogonal, and if they annihilate a real gaugino they lie in the complexification of a real totally isotropic subspace, whose dimension cannot exceed $\min(s,t)$ (Lemmas 2.2 and 2.3). Moving timelike directions into space lowers this number, and with it the number of null directions that can be projected out. What remains after the projection is a theory on the orthogonal complement of the null frame, of dimension $D'=s+t-2k$ for k null vectors. We show that the cubic terms can be absorbed only if the residual spinors satisfy (1.1) themselves, and this forces $D'=10$ whenever $D'>2$. The main result, proved in Sec. 5.4, reads as follows.

Main Theorem. *Let a supersymmetric Yang–Mills theory with a Majorana–Weyl gaugino in signature (s,t) be built on a frame of k null vectors in the manner of Nishino, in the sense of the axioms (N1)–(N5) of Definition 5.3. Then $k\le\min(s,t)$ and $s+t-2k\in\{2,10\}$, so that $s-t\in\{8,0,-8\}$. In particular there is no such theory in any of the signatures* (1.2) *with $n\ge1$, and none in $27+3$ or in $19+3$.*

Figure 1 shows the geometry. Nishino's signatures lie on the plane $s+t-2k=10$, while every signature of the families (1.2) with $n\ge1$ has residual dimension at least 16 for every admissible k. The necessary residual-dimension condition depends on $s-t$ and the available Witt index, beyond their residue data.[1] The conditions of the Main Theorem are sharp for residual dimension ten. A complex isometry that keeps D and the rank of the frame carries Nishino's equations to every signature with $s+t-2k=10$ and $s-t\in\{8,0,-8\}$, and the transported theory keeps a Majorana–Weyl gaugino and satisfies (N1)–(N5) (Theorem 5.12). For residual dimension at least three a theory therefore exists exactly when the Main Theorem allows it (Corollary 5.13). In particular 9+9 and 9+17, two signatures of the second degenerate family of Ref. [23], do carry such theories, while its members $9+(1+8n)$ with $n\ge3$ do not. The conditions are sharp for residual dimension two as well. In $(k+1)+(k+1)$ Nishino's transformations, written with a real frame of rank k, satisfy (N1)–(N5): the cubic gaugino terms vanish identically, and the supersymmetry algebra closes off shell on translations along

[1] Bott periodicity makes the real Clifford algebras of $\mathbb{R}^{s,t}$ and $\mathbb{R}^{s+4,t-4}$ isomorphic, but no isomorphism carries the space of generators $\mathbb{R}^{s,t}$ onto $\mathbb{R}^{s+4,t-4}$. Null vectors, their orthogonality and the bound $\min(s,t)$ are properties of the generators.

one null direction of the residual plane (Theorem 5.14). The Main Theorem is therefore a classification. A theory of rank k exists if and only if $s+t-2k=10$ and $s-t\in\{8,0,-8\}$, or $s+t-2k=2$ and $s=t$ (Corollary 5.15).

The exclusion of the families (1.2) does not depend on the null contraction in the translation bracket either. Whatever fixed tensor contracts the source in the field equation of the gauge field, with or without a constraint on the gaugino, the cubic gaugino terms cannot be absorbed once the residual dimension is sixteen or more, and every real frame in these families leaves at least sixteen (Theorem 7.3 and Corollary 7.4). This covers, for instance, the contraction of the eleven-form charge of $27+3$ with ten spacelike vectors. Several gauginos do not help either. If each gaugino transforms into the field strength with its own parameter, as in extended supersymmetry, the cubic terms that couple two different gauginos must vanish, and they do not once the residual dimension is three or more (Proposition 7.5).

These results do not exclude every theory in $27+3$, and no statement about the superalgebra alone could, since the superalgebra acts on superfields (Remark 7.6). An exclusion has to name the fields and their transformation rules. Theories with auxiliary fields, with transformation rules nonlinear in the gaugino, with gauginos mixed by their transformation rules, or with a gauge structure that is not a Lie algebra lie outside the present results.

For a single time we can say more, without any assumption about how the theory is built. In $(9+8n)+1$ the anticommutator of two supercharges contains the momentum, and in a unitary theory the supercharges that act nontrivially on a massless state generate a Clifford algebra with 2^{3+4n} generators. A massless multiplet without tensor charges then has at least $2^{2^{2+4n}}$ states, half of them bosonic (Theorem 6.1). For $n=1$ that is 2^{64} states, against the 16 bosonic states of a vector field in eighteen dimensions.

The paper is organized as follows. Section 2 collects what we need about spinors in signature (s,t): the decomposition of the symmetric square of a Weyl spinor, which shows that the superalgebras of Ref. [23] are the most general ones and carry no information about dynamics, and two lemmas on isotropic subspaces. Section 3 proves the theorem on which everything else rests: if symmetric gamma matrices obey an identity of the type that absorption requires, the spinor dimension is $2(d-2)$. Section 4 applies it to ordinary super Yang–Mills theory. Section 5 describes null frames by fermionic oscillators, states the axioms, proves the reduction to ten residual dimensions and the Main Theorem, and shows that its conditions are also sufficient. Section 6 treats the single-time family by representation theory, and Sec. 7 spells out what the results mean for $27+3$ and $19+3$, including the transfer of Nishino's construction (Proposition 7.1), arbitrary contractions of the source (Theorem 7.3), several gauginos (Proposition 7.5) and the scope of the axioms. Appendix A proves the oscillator decomposition, Appendix B describes numerical checks in explicit gamma matrices, Appendix C lists the symmetric squares for the families (1.2), Appendix D checks Proposition 7.1 in explicit gamma matrices, Appendix E does the same for Theorems 5.12 and 7.3, and Appendix F for Theorem 5.14 and Proposition 7.5. Figure 2 shows how the results depend on one another.

2. Spinors in signature (s,t)

2.1. **Conventions.** Let $V=\mathbb{R}^{s,t}$ with metric

$$\eta=\operatorname{diag}(+1,\dots,+1,-1,\dots,-1),$$

with s plus signs and t minus signs, and $D=s+t$. Except in Remark 5.11 we take $D=2r$ with $r\geq 1$. The Clifford relations are $\gamma^\mu\gamma^\nu+\gamma^\nu\gamma^\mu=2\eta^{\mu\nu}$. The Dirac module Δ of the complex Clifford algebra $\mathrm{Cl}(V_{\mathbb{C}})$ has dimension 2^r and splits into the Weyl modules $\Delta_\pm$ of dimension 2^{r-1}, the eigenspaces of the chirality operator Γ_*, normalized so that $\Gamma_*^2=1$. For a vector u we write $\not{u}=u_\mu\gamma^\mu$, and $\gamma^{\mu_1\cdots\mu_p}$ is the antisymmetrized product of unit weight, so that $\gamma^{\mu_1\cdots\mu_p}=\gamma^{\mu_1}\cdots\gamma^{\mu_p}$ when the indices are distinct and the basis is orthonormal.

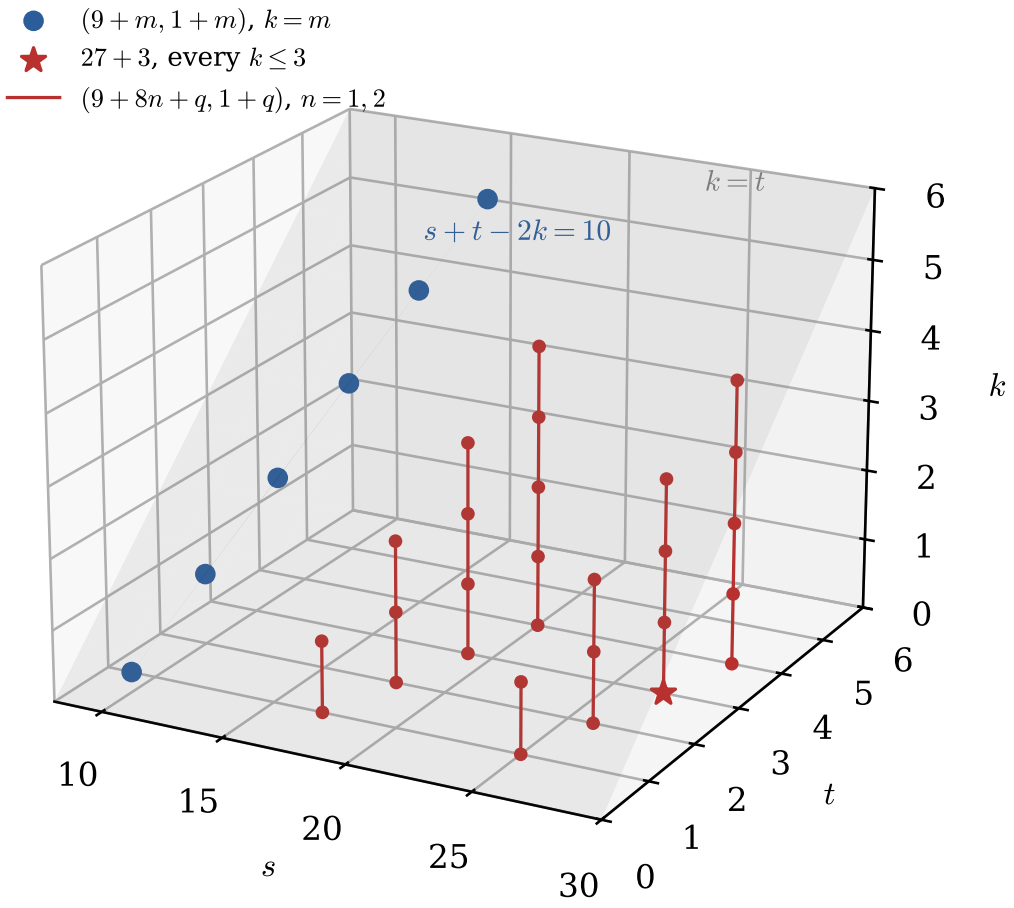


FIGURE 1. Signatures (s,t) and numbers k of null vectors. Blue points: Nishino's signatures $(9+m)+(1+m)$ with $k=m$, on the plane $s+t-2k=10$ (blue). Red segments: the signatures $(9+8n+q)+(1+q)$ of the families (1.2) for $n=1,2$ and $q=0,1,2,3$, with every $k\le t$; the star marks $27+3$. The grey surface is the wall $k=t$, beyond which real null frames do not exist. No red point reaches the blue plane.

There is a nondegenerate bilinear form B on Δ, unique up to a factor once a sign $\theta=\pm1$ is chosen, such that

$$B(\gamma^\mu\psi,\chi)=\theta\,B(\psi,\gamma^\mu\chi),\qquad B(\psi,\chi)=\epsilon_B\,B(\chi,\psi), \tag{2.1}$$

with $\epsilon_B=\pm1$ determined by r and θ [28, 29, 30]. We write $\bar\psi\chi=B(\psi,\chi)$ and $\bar\psi\gamma^{\mu_1\cdots\mu_p}\chi=B(\psi,\gamma^{\mu_1\cdots\mu_p}\chi)$. The form B pairs Δ_+ with Δ_+ when r is even and with Δ_- when r is odd.

A real structure on Δ is an antilinear map $C:\Delta\to\Delta$ with $C^2=1$ that commutes with every γ^μ. It exists exactly when $s-t\equiv0,1,2 \pmod 8$, and it preserves $\Delta_\pm$ exactly when $s-t\equiv0 \pmod 8$ [31, 28, 29]. A Majorana spinor is a spinor with $C\psi=\psi$. Majorana–Weyl spinors, which are the spinors of all the theories in this paper, therefore exist if and only if $s-t\equiv0 \pmod 8$, and every signature in (1.2) and in Nishino's list has this property.

2.2. **The symmetric square of a Weyl spinor.** The anticommutator of two supercharges in the Weyl module Δ_+ is a symmetric element of $\Delta_+\otimes\Delta_+$, so the superalgebras that can be written down in signature (s,t) are classified by the decomposition of $\mathrm{Sym}^2\Delta_+$ [34, 30]. We write $\Lambda^p=\Lambda^pV_{\mathbb C}$, and Λ^r stands for the half Λ^r_+ of the middle-dimensional forms selected by Δ_+.

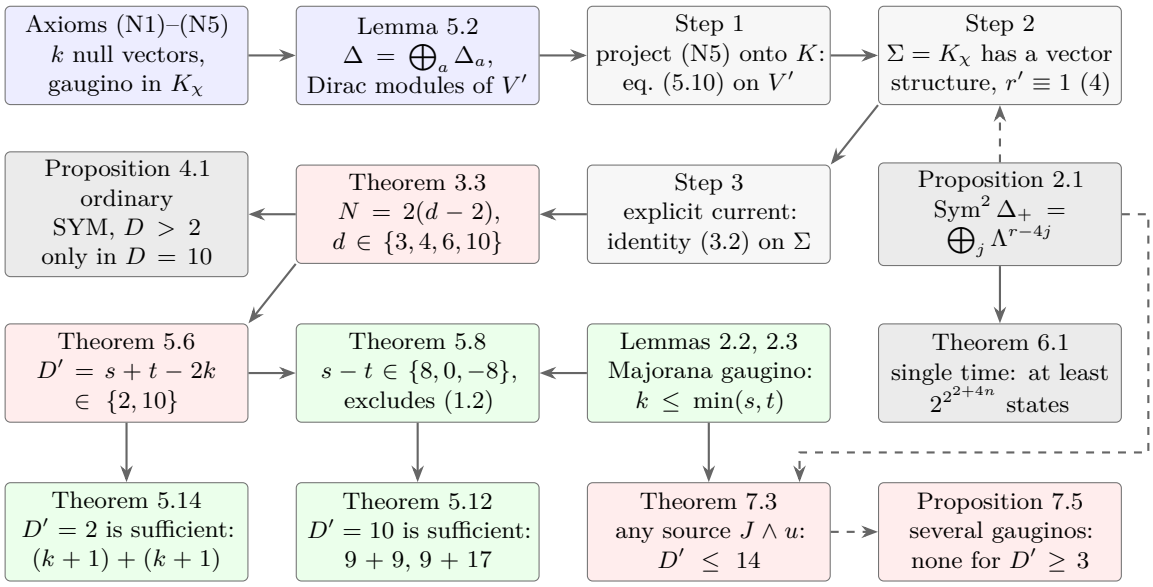


FIGURE 2. The logical structure of the paper. The axioms pass through the oscillator decomposition and Steps 1–3 to the identity (3.2) on the residual spinors; Theorem 3.3 turns that identity into $D' \in \{2, 10\}$, and the bound on the number of null vectors converts this into a condition on $s-t$. Proposition 2.1 enters Step 2 (dashed) and the single-time theorem. Proposition 4.1 is the case without null vectors. In the bottom row, Theorems 5.14 and 5.12 show that the conditions of the Main Theorem are sufficient when $D' = 2$ and $D' = 10$. Theorem 7.3 drops (N4) and allows any contraction of the source; it uses Proposition 2.1 (dashed) and, for real gauginos, the bound on the number of null vectors. Proposition 7.5 uses the same projection (dashed) and excludes several gauginos.

Proposition 2.1. *For $D = 2r$,*

$$\mathrm{Sym}^2 \Delta_+ \cong \bigoplus_{j\ge 0,\ r-4j\ge 0} \Lambda^{r-4j}, \qquad \Lambda^2 \Delta_+ \cong \bigoplus_{j\ge 0,\ r-2-4j\ge 0} \Lambda^{r-2-4j}. \tag{2.2}$$

In particular $\Lambda^1 \subset \mathrm{Sym}^2 \Delta_+$ if and only if $r \equiv 1 \pmod 4$, that is $D \equiv 2 \pmod 8$, and $\Lambda^1 \subset \Lambda^2 \Delta_+$ if and only if $r \equiv 3 \pmod 4$.

Proof. The forms $(\psi, \chi) \mapsto \bar\psi \gamma^{\mu_1 \cdots \mu_p} \chi$ restricted to $\Delta_+ \times \Delta_+$ vanish unless $p \equiv r \pmod 2$, and those with $p \equiv r \pmod 2$, $0 \le p \le r$, give an isomorphism $\Delta_+ \otimes \Delta_+ \cong \bigoplus_p \Lambda^p$ [28, 29]. (As a check, $\sum_{p\equiv r\,(2),\, p<r} \binom{2r}{p} + \frac{1}{2}\binom{2r}{r} = 2^{2r-2} = (\dim \Delta_+)^2$.) It remains to decide which summands are symmetric. By (2.1),

$$\bar\psi \gamma^{\mu_1 \cdots \mu_p} \chi = \theta^p\, B(\gamma^{\mu_p \cdots \mu_1} \psi, \chi) = \theta^p (-1)^{p(p-1)/2} \epsilon_B\, \bar\chi \gamma^{\mu_1 \cdots \mu_p} \psi,$$

so the p-th form has symmetry $\sigma_p = \epsilon_B \theta^p (-1)^{p(p-1)/2}$, and $\sigma_{p+2}/\sigma_p = \theta^2 (-1)^{2p+1} = -1$. The symmetric summands are therefore either those with $p \equiv r \pmod 4$ or those with $p \equiv r - 2 \pmod 4$. Using $\binom{2r}{p} = \binom{2r}{2r-p}$ and summing over the fourth roots of unity, the first set has total dimension

$$\frac{1}{2} \sum_{\substack{0\le p\le 2r \\ p\equiv r\,(4)}} \binom{2r}{p} = \frac{1}{8} \sum_{\omega^4 = 1} \omega^{-r} (1+\omega)^{2r} = \frac{1}{8}(2^{2r} + 2^{r+1}) = 2^{2r-3} + 2^{r-2},$$

because $i^{-r}(1+i)^{2r} = i^r (1-i)^{2r} = 2^r$ and the term with $\omega = -1$ vanishes. This is $N(N+1)/2$ for $N = 2^{r-1}$, while the second set has dimension $N(N-1)/2$. Hence the symmetric summands are those with $p \equiv r \pmod 4$. □

Appendix C lists the decomposition for the families (1.2). For $(9+8n)+1$ one has $r = 5+4n$, and $\mathrm{Sym}^2 \Delta_+$ consists of the forms of degrees $1, 5, \ldots, 5+4n$; for $(10+8n)+2$ the degrees are $2, 6, \ldots, 6+4n$; for $(11+8n)+3$ they are $3, 7, \ldots, 7+4n$; and for $(12+8n)+4$ they are $0, 4, \ldots, 8+4n$. These are exactly the right-hand sides of eqs. (3.5)–(3.8) of Ref. [23]. The superalgebras proposed there are therefore maximal central supertranslation algebras: the anticommutator of two supercharges is an arbitrary element of $\mathrm{Sym}^2 \Delta_+$, with a central charge for every summand. Such an algebra exists for every Weyl spinor in every signature, and its existence alone does not establish a field-theoretic realization.[2] It contains the momentum only for $r \equiv 1 \pmod 4$.

2.3. **Isotropic subspaces.** A subspace U of V or of $V_{\mathbb{C}}$ is totally isotropic if $u \cdot u' = 0$ for all $u, u' \in U$.

Lemma 2.2. *A real totally isotropic subspace $U \subset \mathbb{R}^{s,t}$ has $\dim U \leq \min(s,t)$, and the bound is attained.*[3]

Proof. Write $V = V_+ \oplus V_-$ with V_+ positive definite of dimension s and V_- negative definite of dimension t. If $u \in U \cap V_-$, then $u \cdot u = 0$ forces $u = 0$, so the projection $U \to V_+$ is injective and $\dim U \leq s$. Exchanging the roles gives $\dim U \leq t$. With orthonormal bases $e_1, \ldots, e_s$ of V_+ and $f_1, \ldots, f_t$ of V_-, the vectors $e_i + f_i$, $i \leq \min(s,t)$, span a totally isotropic subspace of dimension $\min(s,t)$. □

The next lemma says that a Majorana spinor cannot distinguish a complex null vector from its complex conjugate.

Lemma 2.3. *Let $s - t \equiv 0, 1, 2 \pmod 8$ and let $\lambda \neq 0$ be a Majorana spinor. The set $U_\lambda = \{u \in V_{\mathbb{C}} : \not{u}\lambda = 0\}$ is a complex totally isotropic subspace invariant under complex conjugation. Hence $U_\lambda = U_{\mathbb{R}} \otimes \mathbb{C}$ with $U_{\mathbb{R}} \subset V$ real and totally isotropic, and $\dim_{\mathbb{C}} U_\lambda \leq \min(s,t)$.*

Proof. If $\not{u}\lambda = 0$ and $\not{u}'\lambda = 0$, then $2(u \cdot u')\lambda = (\not{u}\not{u}' + \not{u}'\not{u})\lambda = 0$, so $u \cdot u' = 0$. Since C is antilinear and commutes with the γ^μ, $C(\not{u}\lambda) = \bar{u}_\mu \gamma^\mu C\lambda = \bar{\not{u}}\lambda$, so $\not{u}\lambda = 0$ implies $\bar{\not{u}}\lambda = 0$. A conjugation-invariant complex subspace is the complexification of its real points, and the real points of a totally isotropic subspace form a totally isotropic subspace of V. Lemma 2.2 finishes the proof. □

Figure 3 illustrates the lemma in $\mathbb{R}^{2,1}$, where $\min(s,t) = 1$: two independent real null vectors are never orthogonal.

3. Vector structures and the cubic identity

Every condition on cubic fermion terms that appears in this paper is a statement about the following structure.

Definition 3.1. Let W be a complex vector space of dimension d with a nondegenerate symmetric bilinear form η, and let Σ be a complex vector space of dimension $N \geq 1$ with dual Σ^*. A *vector structure* on Σ is a pair of linear maps, γ from W to the symmetric maps $\Sigma \to \Sigma^*$ and $\tilde{\gamma}$ from W to the maps $\Sigma^* \to \Sigma$, such that in a basis of W

$$\tilde{\gamma}^\mu \gamma^\nu + \tilde{\gamma}^\nu \gamma^\mu = 2\eta^{\mu\nu}\, 1_\Sigma, \qquad \gamma^\mu \tilde{\gamma}^\nu + \gamma^\nu \tilde{\gamma}^\mu = 2\eta^{\mu\nu}\, 1_{\Sigma^*}. \tag{3.1}$$

In a basis e_α of Σ we write $\gamma^\mu_{\alpha\beta}$ and $\tilde{\gamma}^{\mu\,\alpha\beta}$ for the matrix entries; $\gamma^\mu_{\alpha\beta} = \gamma^\mu_{\beta\alpha}$.

[2] For $27+3$ one has $r = 15$ and $\mathrm{Sym}^2 \Delta_+ = \Lambda^{15}_+ \oplus \Lambda^{11} \oplus \Lambda^7 \oplus \Lambda^3$, of dimension $77\,558\,760 + 54\,627\,300 + 2\,035\,800 + 4\,060 = 134\,225\,920 = 2^{13}(2^{14}+1)$. These are the 3-, 7-, 11- and 15-form charges of eq. (2.1) of Ref. [24].

[3] The number $\min(s,t)$ is the Witt index of η [35, 36].

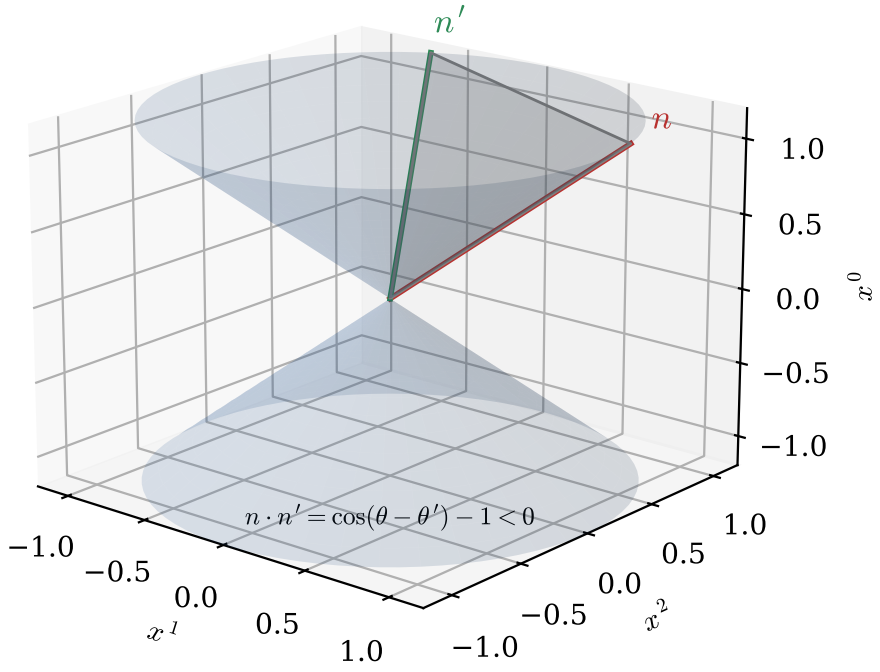


FIGURE 3. The light cone of $\mathbb{R}^{2,1}$. Two independent real null vectors $n = (\cos\theta, \sin\theta, 1)$ and $n' = (\cos\theta', \sin\theta', 1)$ satisfy $n \cdot n' = \cos(\theta - \theta') - 1 < 0$, so the plane they span (grey) is not isotropic. In $\mathbb{R}^{s,t}$ the largest isotropic subspaces have dimension $\min(s, t)$ (Lemma 2.2).

The standard example is $\Sigma = \Delta_+$ in $D = 2r$ with r odd, $\Sigma^* \cong \Delta_-$ through B, $\gamma^\mu_{\alpha\beta} = B(e_\alpha, \gamma^\mu e_\beta)$, and $\tilde\gamma^\mu$ the Clifford action $\Delta_- \to \Delta_+$; by Proposition 2.1 the forms are symmetric when $r \equiv 1 \pmod 4$. In ten dimensions these are the familiar 16×16 matrices $\sigma^\mu_{\alpha\beta}$ and $\bar\sigma^{\mu\,\alpha\beta}$ [37].

Lemma 3.2. *Let $(\Sigma, \gamma, \tilde\gamma)$ be a vector structure with $d \geq 3$. Then*

(a) each $\tilde\gamma^\mu$ is symmetric, and in an orthonormal basis $\tilde\gamma^\mu = \eta^{\mu\mu}(\gamma^\mu)^{-1}$ (no sum);
(b) $\gamma_\mu \tilde\gamma^\mu = d\, 1_{\Sigma^*}$ and $\gamma_\mu \tilde\gamma^\nu \gamma^\mu = -(d-2)\gamma^\nu$;
(c) $\operatorname{tr}_{\Sigma^*}(\gamma^\mu \tilde\gamma^\nu) = \gamma^\mu_{\alpha\beta} \tilde\gamma^{\nu\,\beta\alpha} = N\eta^{\mu\nu}$.

Proof. (a) Setting $\mu = \nu$ in (3.1) gives $\gamma^\mu \tilde\gamma^\mu = \eta^{\mu\mu} 1_{\Sigma^*}$ and $\tilde\gamma^\mu \gamma^\mu = \eta^{\mu\mu} 1_\Sigma$, and the inverse of a symmetric map is symmetric. (b) Contracting the second relation in (3.1) gives the first identity, and then $\gamma_\mu \tilde\gamma^\nu \gamma^\mu = \gamma_\mu(2\eta^{\nu\mu} - \tilde\gamma^\mu \gamma^\nu) = 2\gamma^\nu - d\gamma^\nu$. (c) On $\Sigma \oplus \Sigma^*$ the operators $\Gamma^\mu = \left(\begin{smallmatrix} 0 & \tilde\gamma^\mu \\ \gamma^\mu & 0 \end{smallmatrix}\right)$ obey $\Gamma^\mu\Gamma^\nu + \Gamma^\nu\Gamma^\mu = 2\eta^{\mu\nu}$, and $\Gamma_* = \operatorname{diag}(1, -1)$ anticommutes with every Γ^μ. In an orthonormal basis take $\mu \neq \nu$ and choose $\rho \notin \{\mu, \nu\}$, which is possible because $d \geq 3$. The invertible operator Γ^ρ commutes with $\Gamma^\mu\Gamma^\nu$ and anticommutes with $\Gamma_*\Gamma^\mu\Gamma^\nu$, so $\operatorname{tr}(\Gamma_*\Gamma^\mu\Gamma^\nu) = 0$; and $\operatorname{tr}(\Gamma^\mu\Gamma^\nu) = -\operatorname{tr}(\Gamma^\nu\Gamma^\mu) = 0$. The restriction of $\Gamma^\mu\Gamma^\nu = \operatorname{diag}(\tilde\gamma^\mu\gamma^\nu, \gamma^\mu\tilde\gamma^\nu)$ to Σ^* is $\gamma^\mu\tilde\gamma^\nu$, and its trace $\frac{1}{2}\operatorname{tr}((1 - \Gamma_*)\Gamma^\mu\Gamma^\nu)$ therefore vanishes for $\mu \neq \nu$. For $\mu = \nu$ it is $N\eta^{\mu\mu}$ by (a). □

The condition that absorption of cubic terms will produce is the following. For a constant $j \in \mathbb{C}$ consider

$$\tfrac{1}{2}(\gamma^\mu_{\alpha\beta}\gamma_{\mu\,\gamma\delta} + \gamma^\mu_{\alpha\gamma}\gamma_{\mu\,\beta\delta}) = j\,\gamma^\sigma_{\alpha\delta}\gamma_{\sigma\,\beta\gamma} \qquad \text{for all } \alpha, \beta, \gamma, \delta. \tag{3.2}$$

For $j = -\frac{1}{2}$ this is the identity (1.1).

Theorem 3.3. *Let $(\Sigma, \gamma, \tilde{\gamma})$ be a vector structure with $d \geq 3$, and suppose that (3.2) holds for some $j \in \mathbb{C}$. Then*

$$N = 2(d-2), \qquad j = -\tfrac{1}{2}, \tag{3.3}$$

so the identity is (1.1). Moreover N is divisible by $2^{\lceil d/2 \rceil - 1}$, and therefore $(d, N) \in \{(3,2), (4,4), (6,8), (10,16)\}$.

Proof. Contract (3.2) with $\tilde{\gamma}^{\nu\,\delta\gamma}$, which is symmetric by Lemma 3.2(a), and sum over γ, δ. Inside the bracket on the left, the first term gives $\gamma^{\mu}_{\alpha\beta}\,\mathrm{tr}(\gamma_\mu \tilde{\gamma}^\nu) = N\gamma^{\nu}_{\alpha\beta}$ and the second gives $(\gamma^\mu \tilde{\gamma}^\nu \gamma_\mu)_{\alpha\beta} = -(d-2)\gamma^{\nu}_{\alpha\beta}$, by Lemma 3.2(b),(c). The right-hand side gives $j(\gamma^\sigma \tilde{\gamma}^\nu \gamma_\sigma)_{\alpha\beta} = -j(d-2)\gamma^{\nu}_{\alpha\beta}$. Since $\gamma^\nu \neq 0$,

$$N - (d-2) = -2j(d-2). \tag{3.4}$$

Next contract with $\tilde{\gamma}^{\nu\,\beta\gamma}$ and sum over β, γ. Both terms in the bracket give $(\gamma^\mu \tilde{\gamma}^\nu \gamma_\mu)_{\alpha\delta} = -(d-2)\gamma^{\nu}_{\alpha\delta}$, and the right-hand side gives $j\gamma^{\sigma}_{\alpha\delta}\,\mathrm{tr}(\gamma_\sigma \tilde{\gamma}^\nu) = jN\gamma^{\nu}_{\alpha\delta}$, so

$$jN = -(d-2). \tag{3.5}$$

Substituting $j = -(d-2)/N$ into (3.4) gives $N^2 - (d-2)N - 2(d-2)^2 = 0$, that is $(N - 2(d-2))(N + d - 2) = 0$. As $N + d - 2 > 0$, we get $N = 2(d-2)$ and $j = -\frac{1}{2}$.

For the divisibility, the operators Γ^μ and Γ_* of the proof of Lemma 3.2(c) make $\Sigma \oplus \Sigma^*$ a module over the complex Clifford algebra on $d+1$ generators. This algebra is semisimple and its irreducible modules have dimension $2^{\lfloor (d+1)/2 \rfloor} = 2^{\lceil d/2 \rceil}$, so $2^{\lceil d/2 \rceil}$ divides $2N = 4(d-2)$. For $d = 5, 7, 8, 9$ it does not, and for $d \geq 11$ one has $4(d-2) < 2^{\lceil d/2 \rceil}$. This leaves $d = 3, 4, 6, 10$. □

Remark 3.4. The four solutions are the Majorana spinors of three and four dimensions, the Weyl spinors of six (with their $Sp(1)$ doublet) and the Majorana–Weyl spinors of ten, and in each case (1.1) holds [2, 3, 4]. Theorem 3.3 is a converse that makes no reference to division algebras and allows a general constant j: the dimension count alone fixes both N and j. The constant j is left free because absorbing the cubic terms into a modified field equation, the mechanism of Nishino's construction, produces (3.2) with a constant that is not known in advance (Sec. 5). Figure 4 compares $2(d-2)$ with the smallest admissible N.

4. Ordinary super Yang–Mills theory

Before turning to null projections we record what Theorem 3.3 gives without them. Let the gauge algebra $\mathfrak{g}$ be a compact non-abelian Lie algebra with structure constants f^{IJK}, totally antisymmetric in an orthonormal basis. Let the gaugino λ^I and the parameter ϵ be Grassmann-odd Weyl spinors in Δ_+ (Majorana–Weyl when $s - t \equiv 0 \pmod 8$), and consider

$$\mathcal{L} = -\tfrac{1}{4} F^I_{\mu\nu} F^{I\mu\nu} + \tfrac{c}{2}\,\bar{\lambda}^I \gamma^\mu D_\mu \lambda^I, \qquad \delta A^I_\mu = a\,\bar{\epsilon}\gamma_\mu \lambda^I, \qquad \delta\lambda^I = b\,\gamma^{\mu\nu} F^I_{\mu\nu}\epsilon, \tag{4.1}$$

with $D_\mu \lambda^I = \partial_\mu \lambda^I + g f^{IJK} A^J_\mu \lambda^K$ and nonzero constants a, b, c, g.

Proposition 4.1. *Assume $D > 2$ and that the gaugino term in (4.1) is a nondegenerate kinetic term, rather than zero or a total derivative. If $\delta\mathcal{L}$ is a total derivative, then the forms $\gamma^{\mu}_{\alpha\beta} = B(e_\alpha, \gamma^\mu e_\beta)$ on Δ_+ define a vector structure satisfying (1.1). Consequently $D = 10$, and with Majorana–Weyl spinors the signature is $(9,1)$, $(5,5)$ or $(1,9)$.*

Proof. The kinetic term is nonzero only if B pairs Δ_+ with $\gamma^\mu \Delta_+ = \Delta_-$, that is for r odd, and it is not a total derivative only if the form $\gamma^{\mu}_{\alpha\beta}$ is symmetric: for Grassmann-odd λ and antisymmetric G, $\lambda^\alpha G_{\alpha\beta} \partial \lambda^\beta = \frac{1}{2}\partial(\lambda^\alpha G_{\alpha\beta}\lambda^\beta)$. With $\Sigma = \Delta_+$, $\Sigma^* \cong \Delta_-$ and $\tilde{\gamma}^\mu$ the Clifford action $\Delta_- \to \Delta_+$, the relations (3.1) are those of $\mathrm{Cl}(V)$, so $(\Delta_+, \gamma, \tilde{\gamma})$ is a vector structure with $d = D$ and $N = 2^{r-1}$.

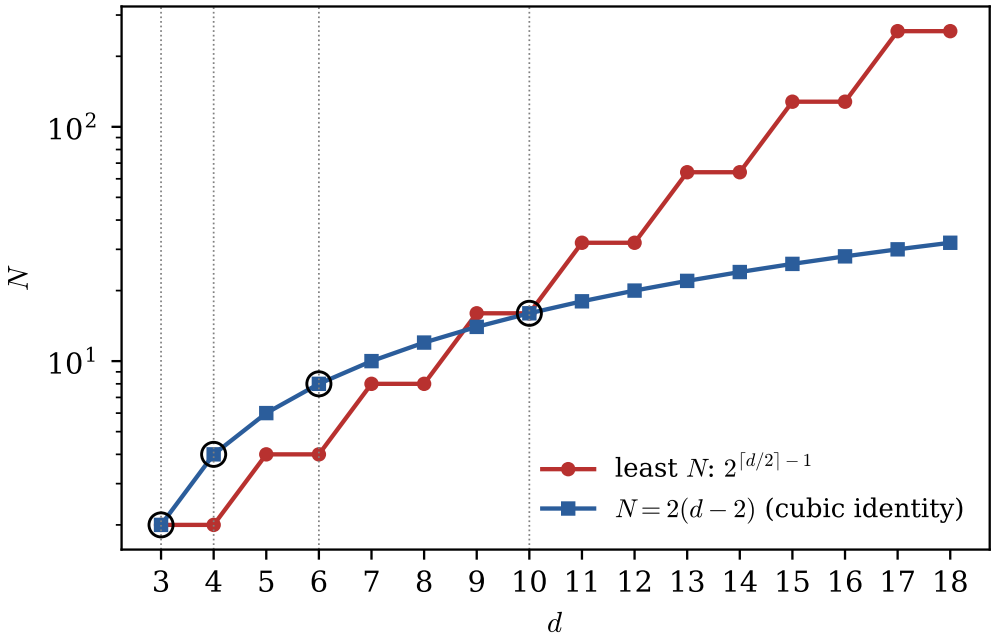


FIGURE 4. The spinor dimension $N = 2(d-2)$ forced by the cubic identity (blue) against the smallest dimension $2^{\lceil d/2 \rceil - 1}$ that a vector structure can have (red). A solution requires the blue value to be a multiple of the red one. This happens for $d = 3, 4, 6, 10$ (circles) and never again, because the red curve grows exponentially.

The terms of $\delta\mathcal{L}$ cubic in λ come only from δA inside the covariant derivative, since $\delta\lambda$ is linear in F and δF is linear in λ. They are

$$\begin{aligned} \tfrac{c}{2}\, ag\, f^{IJK}\, (\bar\lambda^I \gamma^\mu \lambda^K)(\bar\epsilon \gamma_\mu \lambda^J) &= \pm \tfrac{c}{2}\, ag\, \gamma^\mu_{\alpha\gamma} \gamma_{\mu\,\delta\beta}\, \epsilon^\delta\, T^{\alpha\beta\gamma}, \\ T^{\alpha\beta\gamma} &= f^{IJK} \lambda^{I\alpha} \lambda^{J\beta} \lambda^{K\gamma}. \end{aligned} \tag{4.2}$$

where the sign depends only on the ordering convention for Grassmann variables. They contain no derivative, so they must vanish. The tensor T is totally symmetric: exchanging two of its indices and relabelling the gauge indices produces one sign from f and one from the anticommuting λ. It spans $\mathrm{Sym}^3 \Sigma$. Indeed $\mathfrak{g}$ contains a subalgebra $\mathfrak{su}(2)$, and for λ with values in it $T^{\alpha\beta\gamma} c_{\alpha\beta\gamma} = 6 c_{\alpha\beta\gamma} \lambda^{1\alpha} \lambda^{2\beta} \lambda^{3\gamma}$ for every symmetric c, after normalizing $f^{123} = 1$. Since the monomials $\lambda^{1\alpha} \lambda^{2\beta} \lambda^{3\gamma}$ are linearly independent, only $c = 0$ annihilates every T. Hence, for every δ, the part of $\gamma^\mu_{\alpha\gamma} \gamma_{\mu\,\delta\beta}$ symmetric in α, β, γ vanishes. Because each factor is symmetric in its own pair of indices, that symmetrization is one third of the sum over the three ways of pairing δ with α, β or γ, which is (1.1). Theorem 3.3 with $N = 2^{r-1}$ and $d = 2r$ gives $2^{r-1} = 4(r-1)$, whose only solution with $r \geq 2$ is $r = 5$. Majorana–Weyl spinors in $D = 10$ require $s - t \in \{8, 0, -8\}$. □

The proposition is the familiar statement that ten is the largest dimension of super Yang–Mills theory [1], obtained here from the dimension count of Theorem 3.3 rather than from an explicit Fierz rearrangement. For $D > 2$, changing the signature does not evade this conventional nondegenerate-kinetic-term ansatz, because the proof uses only D. The degenerate two-dimensional case is not covered by the proposition. The higher-dimensional constructions examined here therefore constrain the gaugino.

5. Null-projected super Yang–Mills theories

5.1. **Null frames as fermionic oscillators.**

Definition 5.1. A *null frame of rank k* is a set of vectors $n_1, \dots, n_k, m_1, \dots, m_k \in V_{\mathbb{C}}$ with

$$n_i \cdot n_j = 0, \qquad m_i \cdot m_j = 0, \qquad n_i \cdot m_j = \delta_{ij}. \tag{5.1}$$

It is *real* if all the vectors lie in V. Its *residual space* is the orthogonal complement V' of $\operatorname{span}(n_i, m_i)$, of dimension $D' = D - 2k$, and we set $v = n_1 \wedge \cdots \wedge n_k$.

The span of the n_i and m_i is a sum of k hyperbolic planes, so for a real frame V' is real of signature $(s-k, t-k)$. By Lemma 2.2, a real null frame of rank k exists exactly when $k \le \min(s,t)$. In Nishino's notation $n_i = n_{(i)}$ and $m_i = m_{(i)}$; his signatures $(9+m)+(1+m)$ carry real frames of rank m with residual space $\mathbb{R}^{9,1}$.

The relations (5.1) say that the operators

$$b_i = \tfrac{1}{\sqrt{2}} \not{n}_i, \qquad b_i^\dagger = \tfrac{1}{\sqrt{2}} \not{m}_i \tag{5.2}$$

obey the canonical anticommutation relations $\{b_i, b_j^\dagger\} = \delta_{ij}$, $\{b_i, b_j\} = \{b_i^\dagger, b_j^\dagger\} = 0$. (The dagger is notation; no adjoint is implied.) The number operators

$$P_i^\uparrow = b_i^\dagger b_i = \tfrac{1}{2} \not{m}_i \not{n}_i, \qquad P_i^\downarrow = b_i b_i^\dagger = \tfrac{1}{2} \not{n}_i \not{m}_i = 1 - P_i^\uparrow \tag{5.3}$$

are Nishino's projectors [22], and they commute with each other. For $a \in \{0,1\}^k$ let Δ_a be the joint eigenspace on which $P_i^\uparrow = a_i$; mode i is occupied in Δ_a if $a_i = 1$. Write $\mathbf{1} = (1, \dots, 1)$, $\mathbf{0} = (0, \dots, 0)$, $\bar{a} = \mathbf{1} - a$, and e_i for the i-th unit vector of $\{0,1\}^k$. For vectors of V' we use primed indices, $\gamma^{\mu'}$.

Lemma 5.2. *(i) $\Delta = \bigoplus_a \Delta_a$. Each $\gamma^{\mu'}$ preserves every Δ_a, and each Δ_a is a Dirac module of $\mathrm{Cl}(V'_{\mathbb{C}})$, of dimension 2^{r-k}.*

(ii) $\not{m}_i$ maps Δ_a with $a_i = 0$ isomorphically onto Δ_{a+e_i} and annihilates Δ_a with $a_i = 1$; $\not{n}_i$ does the reverse. Hence

$$K := \bigcap_i \ker \not{m}_i = \Delta_{\mathbf{1}}, \qquad \widehat{K} := \bigcap_i \ker \not{n}_i = \Delta_{\mathbf{0}}. \tag{5.4}$$

(iii) $T = \not{m}_1 \not{m}_2 \cdots \not{m}_k$ maps $\widehat{K}$ isomorphically onto K, annihilates Δ_a for $a \neq \mathbf{0}$, and satisfies $T\gamma^{\mu'} = (-1)^k \gamma^{\mu'} T$; in particular T commutes with $\mathrm{Spin}(V')$.

(iv) $B(\Delta_a, \Delta_b) = 0$ unless $b = \bar{a}$.

(v) On each Δ_a the chirality Γ_* equals $\pm(-1)^{|a|}\Gamma'_*$, where Γ'_* is the chirality of V', $|a|$ is the number of occupied modes, and the overall sign does not depend on a. Thus $\Delta_a \cap \Delta_+$ and $\Delta_a \cap \Delta_-$ are the two Weyl modules of V'.

The proof is in Appendix A. Figure 5 shows the case $k = 3$. The kernel K of all the $\not{m}_i$ is the fully occupied corner, the kernel $\widehat{K}$ of all the $\not{n}_i$ is the empty corner, T climbs from one to the other, and B pairs opposite corners.

5.2. **The axioms.**

Definition 5.3. A *null-projected super Yang–Mills theory of rank k* in signature (s,t), $D = 2r$, consists of a null frame of rank k, a compact non-abelian gauge algebra $\mathfrak{g}$, a gauge field A_μ^I, a Grassmann-odd gaugino λ^I and Grassmann-odd supersymmetry parameters ϵ, subject to the following conditions.

(N1) *Constraint.* The complexified gaugino value space is the full corner $K_\chi = K \cap \Delta_\chi$ for a fixed chirality χ; λ^I takes values there, that is, $\not{m}_i \lambda^I = 0$ for all i. The parameter ϵ ranges over a Weyl module $\Delta_{\chi'}$.[4]

[4] As in Nishino's construction, ϵ is not constrained. The proof uses only the component of ϵ in $\widehat{K}$, which by Lemma 5.2(iv) is the only component that enters the residual components of δA. Any family of parameters whose $\widehat{K}$-components form a nonzero $\mathrm{Spin}(V')$-invariant subspace leads to the same conclusions.

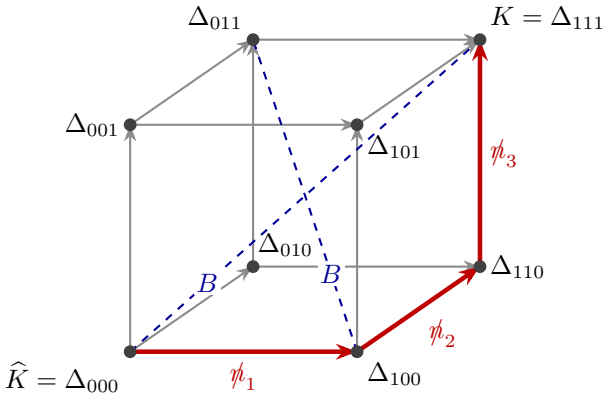


FIGURE 5. The decomposition of the Dirac module for a null frame of rank $k = 3$. The vertices are the joint eigenspaces Δ_a of the number operators $P_i^{\uparrow}$, each a Dirac module of the residual space V'. The gaugino lives in the fully occupied corner K; the supersymmetry parameters that act on the residual components of the gauge field live in the opposite corner $\widehat{K}$.

(N2) *Transformations.* $\delta A_\mu^I = a\,\bar\epsilon\gamma_\mu\lambda^I$ with $a \neq 0$, and $\delta\lambda^I = \Phi(F^I)\epsilon$ for a linear map Φ from two-forms to $\operatorname{End}\Delta$ that does not depend on λ. The covariant derivative is $D_\mu\lambda^I = \partial_\mu\lambda^I + g f^{IJK} A_\mu^J \lambda^K$ with $g \neq 0$.

(N3) *Residual gauge field.* The components $\delta A^I_{\mu'}$ along V' do not vanish identically.

(N4) *Closure.* The commutator of two supersymmetries contains the translation by

$$\xi^\mu = c\,\bar\epsilon_2\gamma^{\mu\nu_1\cdots\nu_k}\epsilon_1\, v_{\nu_1\cdots\nu_k}, \qquad c \neq 0, \tag{5.5}$$

and its components $\xi^{\mu'}$ along V' do not vanish identically. The displayed translation bracket is the coefficient in the supersymmetry anticommutator, so its matrix in the commuting spinor indices is symmetric. This graded symmetry is part of the closure requirement.

(N5) *Absorption.* For every λ with values in $K_\chi \otimes \mathfrak{g}$ there are J^I_σ, depending on λ but not on ϵ, such that for every ϵ the part of $\delta(\gamma^\mu D_\mu\lambda^I)$ quadratic in λ equals

$$\gamma^{\sigma\nu_1\cdots\nu_k}\epsilon\, v_{\nu_1\cdots\nu_k}\, J^I_\sigma. \tag{5.6}$$

Axiom (N5) is what makes the theory consistent: a term of the form (5.6) is cancelled by a source term in the field equation of the gauge field, which becomes $D^\mu F_{\mu[\sigma} v_{\nu_1\cdots\nu_k]} \propto J_{[\sigma} v_{\nu_1\cdots\nu_k]}$. For $k = 0$ the axiom says that the cubic terms have the form $\gamma^\sigma\epsilon J_\sigma$.

Example 5.4. Nishino's theories [22] in $(9+m)+(1+m)$ satisfy (N1)–(N5) with $k = m$. His constraint (2.12), the contraction of v with $\gamma^\mu\lambda$, is equivalent to $\not{p}_{(i)}\lambda = 0$ for every i, and he rewrites it as $\lambda = P^{\uparrow}\lambda$ in eq. (2.22). The transformations (N2) are his eq. (2.3), with $\Phi(F)\epsilon = \frac{1}{4}\gamma^{\mu\nu\rho_1\cdots\rho_m}\epsilon F_{\mu\nu} v_{\rho_1\cdots\rho_m}$, and the closure (N4) is his eqs. (2.16)–(2.18). Axiom (N5) is his eq. (2.29): the surviving cubic term is $\gamma^{[m+1]}\epsilon$ contracted with the bilinear $\bar\lambda^K\gamma_{[m+1]}\lambda^J$, and for gauginos in K this $(m+1)$-form is the wedge product of v with a residual vector (Lemma A.1), so the term has the form (5.6); it is absorbed by the source in his field equation (2.15).[5] The residual space is $\mathbb{R}^{9,1}$, so $D' = 10$.

The variation of the gaugino equation follows directly from the axioms.

Lemma 5.5. *Under (N2), the part of $\delta(\gamma^\mu D_\mu\lambda^I)$ quadratic in λ is $ag\, f^{IJK}\gamma^\mu\lambda^K\,(\bar\epsilon\gamma_\mu\lambda^J)$.*

Proof. $\delta(D_\mu\lambda^I) = D_\mu(\delta\lambda^I) + g f^{IJK}\delta A^J_\mu\lambda^K$. The first term is independent of λ because Φ is, and the second is $agf^{IJK}(\bar\epsilon\gamma_\mu\lambda^J)\lambda^K$. □

[5] Nishino writes $v_{[4n]}$ for the class $m = 4n$ and treats $m = 4n+1, 4n+2, 4n+3$ in the same way. We write k for the number of null vectors in all cases.

5.3. **Reduction to the residual space.**

Theorem 5.6. *In a null-projected super Yang–Mills theory of rank k the residual dimension $D' = D - 2k$ equals* 10 *or* 2.

Proof. D' is even, and $D' \geq 1$ because (N3) requires residual directions. Assume $D' \geq 3$; we prove $D' = 10$. The proof has three steps.

Step 1: projection onto the occupied corner. Since $n_i \cdot m_j = \delta_{ij}$, the inverse metric decomposes as $\eta^{-1} = \eta'^{-1} + \sum_i (n_i \otimes m_i + m_i \otimes n_i)$, so

$$\gamma^{\mu}\lambda\,(\bar\epsilon\gamma_{\mu}\lambda') = \gamma^{\mu'}\lambda\,(\bar\epsilon\gamma_{\mu'}\lambda') + \sum_i \big[\not{n}_i\lambda\,(\bar\epsilon\not{m}_i\lambda') + \not{m}_i\lambda\,(\bar\epsilon\not{n}_i\lambda')\big]. \tag{5.7}$$

For $\lambda, \lambda' \in K$ the bracket vanishes, because $\not{n}_i$ annihilates K. By Lemma 5.5 the left-hand side of (N5) is therefore

$$ag\, f^{IJK}\gamma^{\mu'}\lambda^K(\bar\epsilon\gamma_{\mu'}\lambda^J), \tag{5.8}$$

which lies in the occupied corner $\boldsymbol{\Delta_1}$ because $\gamma^{\mu'}$ preserves every Δ_a. On the right-hand side of (5.6), write $\gamma(\omega)$ for the antisymmetrized product of gamma matrices attached to a $(k+1)$-vector ω, normalized so that $\gamma(u_0 \wedge \cdots \wedge u_k) = \not{u}_0 \cdots \not{u}_k$ for mutually orthogonal u_i. Up to a nonzero factor, which we absorb into J, the right-hand side is $\gamma(J \wedge v)\epsilon$. Decompose $J = J' + \sum_i [(J \cdot n_i)\, m_i + (J \cdot m_i)\, n_i]$ with $J' \in V'_{\mathbb{C}}$. Since $n_i \wedge v = 0$, the vector $u' \in V'_{\mathbb{C}}$ is orthogonal to all the n_l, and m_i is orthogonal to all n_l with $l \neq i$,

$$\gamma(u' \wedge v) = \not{u}'\,T, \qquad \gamma(n_i \wedge v) = 0, \qquad \gamma(m_i \wedge v) = \pm\Big(\prod_{l\neq i}\not{n}_l\Big)(P_i^{\downarrow} - P_i^{\uparrow}), \tag{5.9}$$

where the last formula uses $\frac{1}{2}(\not{n}_i\not{m}_i - \not{m}_i\not{n}_i) = P_i^{\downarrow} - P_i^{\uparrow}$. Now take $\epsilon \in \widehat{K}_{\chi'} = \widehat{K} \cap \Delta_{\chi'}$. Then $\gamma^{\mu'}T\epsilon \in \boldsymbol{\Delta_1}$, while $(\prod_{l\neq i}\not{n}_l)(P_i^{\downarrow} - P_i^{\uparrow})\epsilon$ lies in $\boldsymbol{\Delta_{1-e_i}}$, in which mode i is empty. Applying the projection $\prod_i P_i^{\uparrow}$ onto $\boldsymbol{\Delta_1}$ to both sides of (N5), we obtain for all $\epsilon \in \widehat{K}_{\chi'}$

$$ag\, f^{IJK}\gamma^{\mu'}\lambda^K(\bar\epsilon\gamma_{\mu'}\lambda^J) = \gamma^{\mu'}T\epsilon\; J^I_{\mu'}. \tag{5.10}$$

Step 2: the residual vector structure. By Lemma 5.2(iii), $x = T\epsilon$ ranges over $K_{\chi''}$ with $\chi'' = (-1)^k\chi'$ as ϵ ranges over $\widehat{K}_{\chi'}$. Define

$$\beta(x, y) = B(T^{-1}x, y) \qquad (x \in K_{\chi''},\ y \in \boldsymbol{\Delta_1}). \tag{5.11}$$

By Lemma 5.2(iv) and the nondegeneracy of B, β is a nondegenerate pairing of $K_{\chi''}$ with the Weyl half of $\boldsymbol{\Delta_1}$ that B pairs with $\widehat{K}_{\chi'}$, and it is $\mathrm{Spin}(V')$-invariant by Lemma 5.2(iii). In this notation

$$\delta A^I_{\mu'} = a\,\beta(T\epsilon, \gamma_{\mu'}\lambda^I), \qquad \xi^{\mu'} = c\,\beta(T\epsilon_2, \gamma^{\mu'}T\epsilon_1) \qquad (\epsilon, \epsilon_1, \epsilon_2 \in \widehat{K}_{\chi'}), \tag{5.12}$$

the first because only the $\widehat{K}$-component of ϵ pairs with $\gamma_{\mu'}\lambda \in \boldsymbol{\Delta_1}$, the second by (5.9), with the normalization of v absorbed into c. Both residual bilinears depend only on the $\widehat{K}$ components of their supersymmetry parameters: T kills every other corner and B pairs opposite corners. Their nonvanishing in (N3) and (N4) therefore survives this restriction. Thus $G^{\mu'}(x, y) = \beta(x, \gamma^{\mu'}y)$ is a nonzero equivariant pairing $K_{\chi''} \times K_{\chi''} \to V'_{\mathbb{C}}$ by (N4), and a nonzero equivariant pairing $K_{\chi''} \times K_{\chi} \to V'_{\mathbb{C}}$ by (N3). By Lemma 5.2(v), $K_{\chi''}$ and K_{χ} are Weyl modules of V'. With $D' = 2r'$, Proposition 2.1 and its analogue for opposite chiralities, $\Delta'_+ \otimes \Delta'_- \cong \bigoplus_j \Lambda^{r'-1-2j}$, show that the tensor square of a Weyl module contains Λ^1 only for r' odd, and the product of the two Weyl modules only for r' even. Hence r' is odd and $K_{\chi''}$ and K_{χ} have the same V'-chirality. Both lie in $\boldsymbol{\Delta_1}$, so by Lemma 5.2(v) they coincide: $K_{\chi''} = K_{\chi} =: \Sigma$.

Next, $G^{\mu'}$ is symmetric on Σ. The commutator of two supersymmetries is antisymmetric under $\epsilon_1 \leftrightarrow \epsilon_2$, and so is the parameter ξ of the translation in it. For Grassmann-odd $x_i = T\epsilon_i$ one has $x_2^{\alpha}G_{\alpha\beta}x_1^{\beta} = -x_1^{\beta}G_{\alpha\beta}x_2^{\alpha}$, so in (5.12) the symmetric part of G contributes a term that is odd under the exchange and the antisymmetric part a term that is even. The even term

must vanish, and since x_1, x_2 are independent Grassmann variables the antisymmetric part of G is zero.[6] By (N4), $G \neq 0$. So G is a nonzero symmetric equivariant map $\mathrm{Sym}^2 \Sigma \to V'_{\mathbb{C}}$, and Proposition 2.1 gives $r' \equiv 1 \pmod 4$. Let Σ_{opp} be the opposite Weyl half of $\Delta_{\mathbf{1}}$. The nondegenerate pairing defines an isomorphism

$$\iota : \Sigma_{\mathrm{opp}} \longrightarrow \Sigma^*, \qquad \iota(z)(x) = \beta(x, z).$$

Writing $\gamma^{\mu'}_{\mathrm{amb}}$ for the ambient Clifford action, define

$$\gamma^{\mu'} = \iota \circ \gamma^{\mu'}_{\mathrm{amb}}|_{\Sigma}, \qquad \tilde{\gamma}^{\mu'} = \gamma^{\mu'}_{\mathrm{amb}}|_{\Sigma_{\mathrm{opp}}} \circ \iota^{-1}.$$

The matrix of the first map is $G^{\mu'}$, hence symmetric. In both compositions of (3.1), $\iota^{-1}\iota$ cancels and the ambient Clifford relations apply. Thus $(\Sigma, \gamma, \tilde{\gamma})$ is a vector structure on $W = V'_{\mathbb{C}}$ with $d = D'$ and $N = \dim \Sigma = 2^{r'-1}$.

Step 3: the identity. Write $x = T\epsilon$ and $S^{I\beta\delta} = f^{IJK}\lambda^{J\beta}\lambda^{K\delta}$. Exchanging β and δ and relabelling $J \leftrightarrow K$ gives one sign from f and one from the anticommuting λ, so S^I is symmetric. In components, (5.10) reads

$$\kappa\, \gamma^{\mu'}_{\alpha\delta}\gamma_{\mu'\,\gamma\beta}\, x^{\gamma} S^{I\beta\delta} = \gamma^{\mu'}_{\alpha\gamma}\, x^{\gamma} J^I_{\mu'}, \tag{5.13}$$

with $\kappa = \pm ag \neq 0$, the sign fixed by the ordering convention. Since (5.13) holds for all x, it is an identity between the matrices that multiply x^{γ}. Contracting with $\tilde{\gamma}^{\nu\,\gamma\alpha}$ and using Lemma 3.2(c),

$$J^{I\nu} = \frac{\kappa}{N}\, \tilde{\gamma}^{\nu\,\gamma\alpha}\gamma^{\mu'}_{\alpha\delta}\gamma_{\mu'\,\gamma\beta}\, S^{I\beta\delta}. \tag{5.14}$$

The current is determined explicitly, without a multiplicity argument. By the symmetry of $\gamma_{\mu'}$ and Lemma 3.2(b),

$$\tilde{\gamma}^{\nu\,\gamma\alpha}\gamma^{\mu'}_{\alpha\delta}\gamma_{\mu'\,\gamma\beta} = (\gamma_{\mu'}\tilde{\gamma}^{\nu}\gamma^{\mu'})_{\beta\delta} = -(D'-2)\gamma^{\nu}_{\beta\delta}.$$

Consequently

$$J^{I\nu} = \kappa j\, \gamma^{\nu}_{\beta\delta} S^{I\beta\delta}, \qquad j = -\frac{D'-2}{N}.$$

This is a linear function of S^I fixed by the absorption equation itself. Substituting into (5.13),

$$\big(\gamma^{\mu'}_{\alpha\delta}\gamma_{\mu'\,\gamma\beta} - j\, \gamma^{\mu'}_{\alpha\gamma}\gamma_{\mu'\,\beta\delta}\big) S^{I\beta\delta} = 0. \tag{5.15}$$

The tensors S^I span $\mathrm{Sym}^2 \Sigma$: for λ with values in an $\mathfrak{su}(2) \subset \mathfrak{g}$, $S^3 = \lambda^1 \otimes \lambda^2 + \lambda^2 \otimes \lambda^1$ up to normalization, and the monomials $\lambda^{1\beta}\lambda^{2\delta}$ are independent.[7] Hence the part of the bracket in (5.15) symmetric in β, δ vanishes:

$$\tfrac{1}{2}\big(\gamma^{\mu'}_{\alpha\delta}\gamma_{\mu'\,\gamma\beta} + \gamma^{\mu'}_{\alpha\beta}\gamma_{\mu'\,\gamma\delta}\big) = j\, \gamma^{\mu'}_{\alpha\gamma}\gamma_{\mu'\,\beta\delta}.$$

Exchanging the names of γ and δ and using the symmetry of $\gamma_{\mu'}$ gives exactly (3.2). Theorem 3.3 now gives $2^{r'-1} = N = 2(D'-2) = 4(r'-1)$, whose only solution is $r' = 5$, that is $D' = 10$. □

Step 1 is where the transplanted constructions fail. In Nishino's signatures the occupied corner K is a sixteen-dimensional Weyl module of the residual $\mathbb{R}^{9,1}$, and (5.10) holds by the ten-dimensional identity (1.1): contracting (1.1) with a symmetric $S^{\beta\delta}$ gives $\gamma^{\mu'}_{\alpha\delta}\gamma_{\mu'\,\gamma\beta}S^{\beta\delta} = -\frac{1}{2}\gamma^{\mu'}_{\alpha\gamma}\gamma_{\mu'\,\beta\delta}S^{\beta\delta}$, so $J_{\mu'}$ is a multiple of the residual fermion current, the source that Nishino places in the field equation (2.15) of the gauge field. Setting the non-residual components of J to zero extends this absorption solution from $\epsilon \in \widehat{K}$ to every parameter: the left-hand side depends only on its $\widehat{K}$ component, and the right-hand side is $\gamma(J \wedge v)\epsilon = \not{J}\, T\epsilon$, which has the same dependence. This solves the absorption condition itself; it supplies no independent

[6] This is the usual reason why the anticommutator of supercharges is symmetric: with Grassmann parameters, $[\delta(\epsilon_1), \delta(\epsilon_2)] = \epsilon_2^{\alpha}\epsilon_1^{\beta}\{Q_\alpha, Q_\beta\}$ up to convention-dependent factors.

[7] For Majorana gauginos the components $\lambda^{J\beta}$ are real Grassmann generators in a real basis of Σ. The identity (5.15) is linear in S^I, so it extends to all complex symmetric tensors.

construction of the transformation Φ or of full closure. In $(9+8n+q)+(1+q)$ with $n \geq 1$ the frame has at most $1+q$ null vectors, so the residual space has dimension at least $s-t = 8+8n \geq 16$, and its Weyl spinors are too large for any identity (3.2).

Remark 5.7. The case $D' = 2$ escapes the argument because Lemma 3.2(c) needs $d \geq 3$. There Σ is one-dimensional, $\gamma^{\mu'}$ is a null vector, and (3.2) holds trivially because both sides vanish. The residual gauge field in two dimensions has no local transverse gauge-boson polarizations; this does not by itself determine the propagation of every other field. By Theorem 5.8(iii) below it can occur only for $s = t$.

5.4. **The Main Theorem.**

Theorem 5.8 (Main Theorem)**.** *Let a null-projected super Yang–Mills theory of rank k in signature (s,t) have a Majorana–Weyl gaugino, so that $s-t \equiv 0 \pmod 8$. Then*

(i) $k \leq \min(s,t)$, and $n_1, \ldots, n_k$ lie in the complexification of a real totally isotropic subspace of V;
(ii) $s+t-2k \in \{2, 10\}$;
(iii) $s-t \in \{8, 0, -8\}$, and $s = t$ if $s+t-2k = 2$.

Proof. (i) The values of a Majorana–Weyl gaugino are real spinors in K_χ, and by (N3) the gaugino does not vanish identically, so K_χ contains a Majorana spinor $\psi \neq 0$. By Lemma 2.3, $U_\psi \supset \operatorname{span}(n_1, \ldots, n_k)$ is the complexification of a real totally isotropic subspace $U_\mathbb{R}$, so $k \leq \dim U_\mathbb{R} \leq \min(s,t)$. (ii) is Theorem 5.6. (iii) From (i) and (ii), $|s-t| = s+t-2\min(s,t) \leq s+t-2k \leq 10$, and $s-t \equiv 0 \pmod 8$ leaves $s-t \in \{8,0,-8\}$; if $s+t-2k = 2$, then $|s-t| \leq 2$ forces $s = t$. □

Corollary 5.9. *There is no null-projected super Yang–Mills theory with a Majorana–Weyl gaugino in any of the signatures* (1.2) *with $n \geq 1$. In particular there is none in $17+1$, $19+3$, $25+1$ or $27+3$.*

Proof. In the families (1.2), $s-t = 8+8n \geq 16$ for $n \geq 1$. □

Remark 5.10. The reality of the gaugino is essential for (i). With a complex Weyl gaugino a complex frame of rank $k > \min(s,t)$ is allowed, and Theorem 5.6 still gives $D' = 10$ when $D' > 2$. For instance $(17,1)$ admits a complex frame of rank four with residual dimension ten, and the absorption equation (5.10) can then be solved, as the numerical check of Appendix B confirms. But the occupied corner of such a frame contains no Majorana spinor (Lemma 2.3), so this projected sector has no Majorana real form compatible with the original real signature and frame. Its residual Clifford algebra is the ten-dimensional complex one. The superalgebras of Ref. [23] are built on Majorana–Weyl generators.

Remark 5.11. For odd D there are no Weyl spinors, so there are no Majorana–Weyl gauginos and the Main Theorem has no odd-dimensional counterpart; the families (1.2) all have even D. Theorem 3.3 itself does not assume that d is even.

Figure 6 summarizes the numerical test of Step 1 in explicit gamma matrices, described in Appendix B: the absorption equation (5.10) has a solution to machine precision in the tested cases with $D' = 10$, and large residuals in the other tested cases.

5.5. **Sufficiency for residual dimension ten.** Theorem 5.8 gives necessary conditions. When the residual dimension is ten they are also sufficient. Nishino's theories carry over to every signature with the same total dimension and the same rank of the frame, as long as the frame stays real, and these are exactly the signatures with residual dimension ten that Theorem 5.8 leaves.

Theorem 5.12. *Let $s+t = 10+2k$ with $k \geq 0$ and $s-t \in \{8,0,-8\}$. Then $\mathbb{R}^{s,t}$ carries a real null frame of rank k with residual space $\mathbb{R}^{9,1}$, $\mathbb{R}^{5,5}$ or $\mathbb{R}^{1,9}$. Nishino's equations for $(9+k)+(1+k)$ (Example 5.4), written with this frame, define a null-projected super Yang–Mills theory of rank k in (s,t) with a Majorana–Weyl gaugino. It satisfies (N1)–(N5), and*

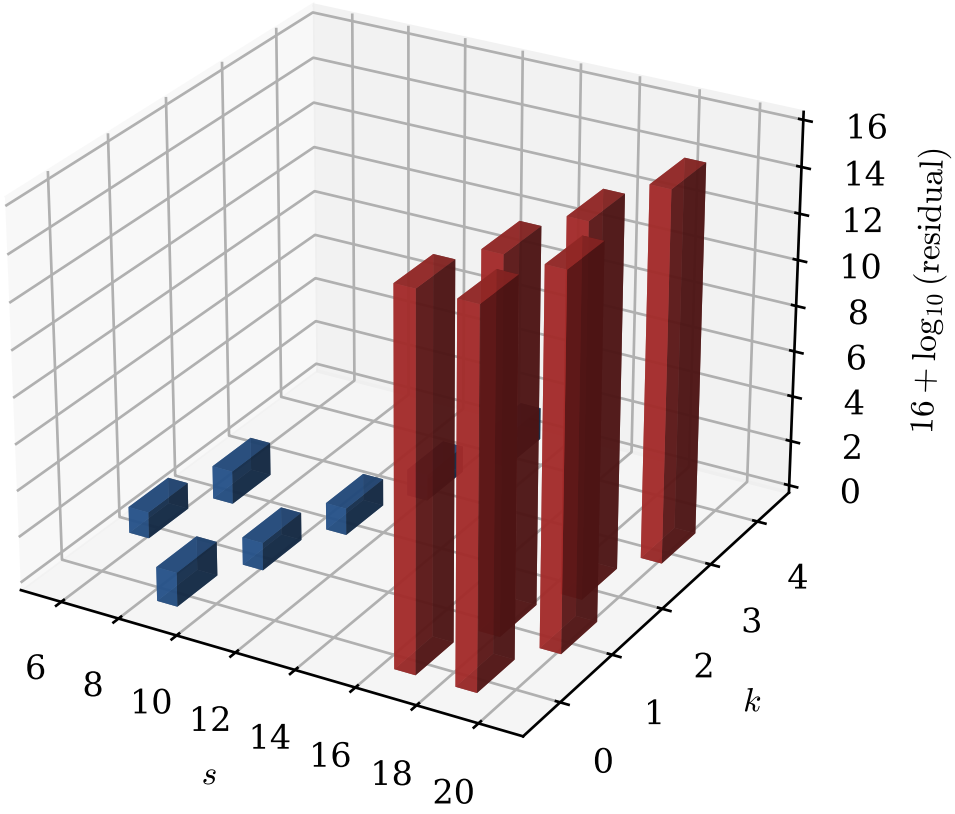


FIGURE 6. Relative residual of the least-squares solution of the absorption equation (5.10) for real frames, plotted as $16 + \log_{10}(\text{residual})$ against s and k. Blue: $(9,1)$, $(10,2)$, $(11,3)$, $(12,4)$, $(13,5)$ with $k = t-1$, and $(6,6)$, $(7,7)$ with $k = 1, 2$, all with $D' = 10$; the residuals are below 3×10^{-15}. Red: $(17,1)$, $(18,2)$, $(19,3)$, $(20,4)$ with every tested k, $D' \in \{16, 18, 20, 22\}$; the residuals lie between 0.97 and 1, so no solution is found in these samples; the exact exclusion follows from the analytic theorem.

its supersymmetry transformations close as in Nishino's eqs. (2.16)–(2.18). For $s - t = 8$ it is Nishino's theory, and for $k = 0$ it is ordinary super Yang–Mills theory in $(9,1)$, $(5,5)$ or $(1,9)$.

Proof. Since $|s-t| \le 8$, $\min(s,t) = \frac{1}{2}(s+t-|s-t|) \ge k+1$, so a real null frame of rank k exists by Lemma 2.2, and its residual space has signature $(s-k, t-k)$, with $s-k+t-k = 10$ and $(s-k)-(t-k) = s-t$.

Let $V_0 = \mathbb{R}^{9+k,1+k}$ with Nishino's frame (n_i^0, m_i^0) and $V = \mathbb{R}^{s,t}$ with a real frame (n_i, m_i). The complexified residual spaces are nondegenerate quadratic spaces of dimension ten, so there is a complex isometry between them, and sending $n_i^0 \mapsto n_i$, $m_i^0 \mapsto m_i$ extends it to a complex isometry $\varphi : V_{0,\mathbb{C}} \to V_{\mathbb{C}}$, by (5.1). It induces an isomorphism of the complex Clifford algebras and, since the Dirac module is unique, a linear isomorphism $\Phi : \Delta^0 \to \Delta$ with $\Phi\,\gamma(u) = \gamma(\varphi u)\,\Phi$. Labelling the chiralities of V so that Φ preserves them, Φ maps Weyl modules to Weyl modules and the corners K, $\widehat{K}$ of the first frame onto those of the second, and it carries B_0 to a nonzero multiple of B, because a form with the property (2.1) for a given θ is unique up to a factor. Nishino's constraint, transformation rules, field equations

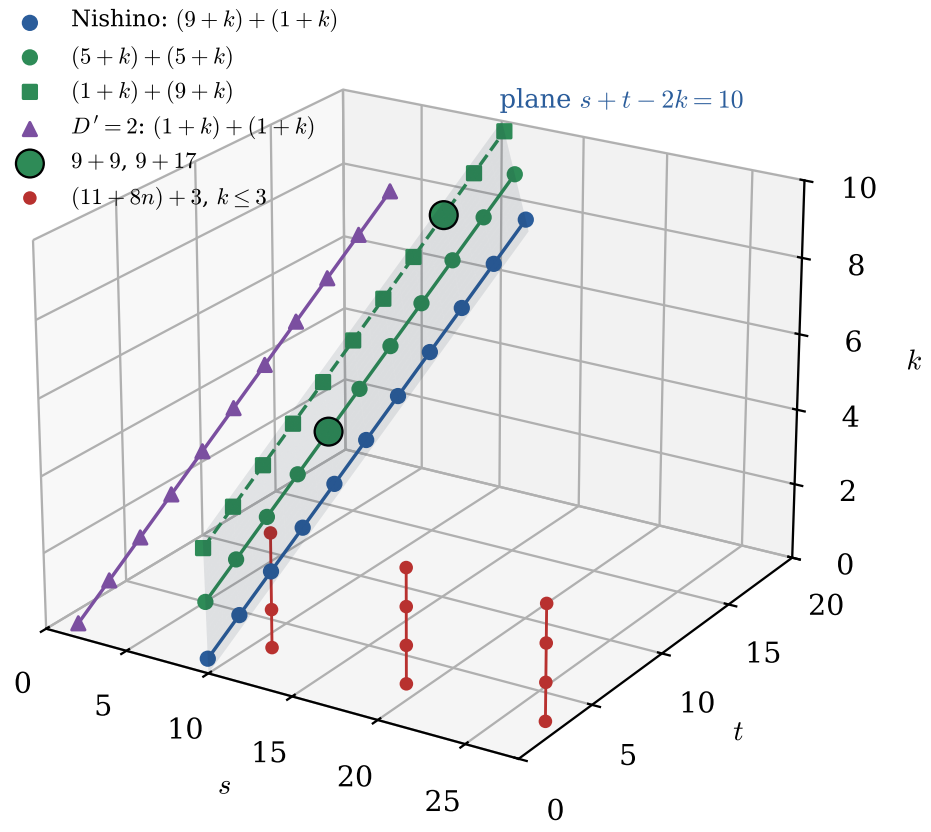


FIGURE 7. The signatures (s,t) and ranks k that carry a null-projected super Yang–Mills theory (Corollary 5.15). Blue: Nishino's signatures $(9+k)+(1+k)$. Green: the signatures $(5+k)+(5+k)$ and $(1+k)+(9+k)$ reached from them by complex isometries that keep D and k (Theorem 5.12); the large green points are $9+9$ and $9+17$ of the second degenerate family. Purple: residual dimension two, $(1+k)+(1+k)$ (Theorem 5.14). Red: the signatures $(11+8n)+3$, $n=0,1,2$, with every rank $k\le 3$; only Sezgin's $11+3$ with $k=2$ meets the plane $s+t-2k=10$.

and closure relations are polynomial identities in the gamma matrices, the metric, the frame, the structure constants and the components of the fields and parameters. The maps φ and Φ preserve all of them. Hence the same equations, written in V with the frame (n_i, m_i), satisfy every identity that they satisfy in V_0, and (N1)–(N5) hold for them over $\mathbb{C}$, with $D'=10$.

It remains to impose reality. Since $s-t\equiv 0 \pmod 8$, Δ carries a real structure C that commutes with every γ^μ and preserves $\Delta_\pm$ (Sec. 2). As the n_i are real, C commutes with every $\not{n}_i$ and preserves K, so the Majorana elements of K_χ form a real form of K_χ; this is (N1) with a Majorana–Weyl gaugino. The form $B'(\psi,\chi)=\overline{B(C\psi,C\chi)}$ again satisfies (2.1) with the same θ, so $B'=cB$ for a constant c, and applying the construction twice gives $|c|=1$. Replacing B by $c^{1/2}B$, we may assume that $B(\psi,\chi)$ is real for Majorana spinors ψ, χ. Then $\delta A = a\,\bar\epsilon\gamma\lambda$ is real for real a, and $\delta\lambda$ is a Majorana spinor because the gamma matrices commute with C and F and v are real. The real fields therefore form a consistent sector of the transported theory. □

Corollary 5.13. *Let $s-t\equiv 0 \pmod 8$ and let a real null frame of rank k have residual dimension $D'=s+t-2k\ge 3$. A null-projected super Yang–Mills theory of rank k with a Majorana–Weyl gaugino exists in (s,t) if and only if $D'=10$ and $s-t\in\{8,0,-8\}$. In*

particular such theories exist in the signatures $9+9$ *(with* $k=4$*) and* $9+17$ *(with* $k=8$*) of the second degenerate family* $9+(1+8n)$ *of Ref.* [23]*, and in no member of that family with* $n \geq 3$*.*

Proof. Necessity is Theorem 5.8, since $D' \geq 3$ excludes $D'=2$. Sufficiency is Theorem 5.12. In $9+(1+8n)$ one has $s-t = 8-8n$, which lies in $\{0,-8\}$ for $n=1,2$, and $D'=10$ gives $k=(s+t-10)/2$. □

The theorem shows what a transfer of Nishino's construction must keep. A complex isometry keeps the total dimension D and the rank k, and the transported theory stays real when the frame can be chosen real, $k \leq \min(s,t)$, and Majorana–Weyl spinors exist, $s-t \equiv 0 \pmod 8$. With $D'=10$ these conditions force $s-t \in \{8,0,-8\}$. In thirty dimensions, for example, Nishino's rank $k=10$ in $19+11$ carries over to $15+15$ and $11+19$, and to no signature with three times. Figure 7 shows the signatures that carry these theories. For residual dimension two, which Theorem 5.8 allows when $s=t$, no ten-dimensional theory can be transported, and Theorem 5.14 below constructs the theory directly.

5.6. **Residual dimension two.** The Main Theorem leaves a second case, $s+t-2k=2$ with $s=t$, so the signatures $(k+1)+(k+1)$ with a frame of rank k. Here no ten-dimensional theory can be transported, and we build the theory directly. The residual plane $V' \cong \mathbb{R}^{1,1}$ is spanned by two real null vectors ℓ_+, ℓ_- with $\eta(\ell_+,\ell_-)=1$. Since $\gamma(\ell_\pm)^2=0$ and $\gamma(\ell_+)\gamma(\ell_-)+\gamma(\ell_-)\gamma(\ell_+)=2$, the two Weyl modules of a Dirac module of $\mathrm{Cl}(V'_{\mathbb{C}})$ are $\ker\gamma(\ell_+)$ and $\ker\gamma(\ell_-)$, and each is one-dimensional. We label $\ell_\pm$ so that the gaugino space $\Sigma = K_\chi$ is $\ker\gamma(\ell_+)$ (Lemma 5.2(i),(v)), and we take ϵ of the chirality χ' for which $T\widehat{K}_{\chi'} = \Sigma$, so that $x = T\epsilon$ ranges over Σ as ϵ ranges over $\widehat{K}_{\chi'}$ (Lemma 5.2(iii)). The pairing β of (5.11) is $\mathrm{Spin}(V')$-invariant and nondegenerate, and $\mathrm{Spin}(1,1)$ acts on the two Weyl modules with opposite weights, so β pairs Σ with $\gamma(\ell_-)\Sigma = \ker\gamma(\ell_-)$; hence $b = \beta(e,\gamma(\ell_-)e) \neq 0$ for a basis vector e of Σ.

Theorem 5.14. *Let* $k \geq 0$*, let* $n_1,\dots,n_k$ *be a real null frame of rank* k *in* $\mathbb{R}^{k+1,k+1}$ *with* $v = n_1 \wedge \cdots \wedge n_k$*, and let* $\mathfrak{g}$ *be a compact non-abelian Lie algebra. Take Nishino's transformations*

$$\delta A^I_\mu = a\,\bar\epsilon\gamma_\mu\lambda^I, \qquad \delta\lambda^I = \tfrac{1}{4}\gamma^{\mu\nu\rho_1\cdots\rho_k}\epsilon\,F^I_{\mu\nu}v_{\rho_1\cdots\rho_k}, \tag{5.16}$$

with $a \neq 0$*, a Majorana–Weyl gaugino with values in* $K_\chi = \Sigma$*, the constraints* $n_i^\mu F^I_{\mu\nu} = 0$ *and* $n_i^\mu D_\mu\lambda^I = 0$ *of Nishino's eqs. (2.11)–(2.13), and the gaugino equation* $\gamma^\mu D_\mu\lambda^I = 0$*. Then:*

- *(i) the constraints are preserved by* (5.16)*, and (N1)–(N5) hold with residual space* $\mathbb{R}^{1,1}$*;*
- *(ii) the part of* $\delta(\gamma^\mu D_\mu\lambda^I)$ *quadratic in* λ *vanishes for every* ϵ*, so (N5) holds with* $J=0$*;*
- *(iii) the commutator of two supersymmetries is, off shell, the translation by* ξ *on* λ*, and the translation by* ξ*, a gauge transformation and a shift of the components* $m_i^\mu A_\mu$ *on* A*;*
- *(iv) the gaugino equation is* $D_{\ell_+}\lambda^I = 0$*, and* ξ *is a multiple of* ℓ_+ *plus a combination of the* n_i*.*

For $k=0$ *these are the transformations of two-dimensional super Yang–Mills theory with one Majorana–Weyl supercharge.*

Proof. A real null frame of rank k exists because $k < \min(s,t) = k+1$, its residual space has signature $(1,1)$, and Majorana–Weyl spinors exist because $s-t=0$. Reality of the transformations follows as in the proof of Theorem 5.12, since the frame is real.

Reduction to the residual plane. The constraint $n_i^\mu F_{\mu\nu}=0$ says that F lies in Λ^2 of $W = \mathrm{span}(n_1,\dots,n_k)\oplus V'$. Raising its indices, the legs along W that are not residual are multiples of the n_i, and these die in the wedge product with v. Hence $\gamma^{\mu\nu\rho_1\cdots\rho_k}F_{\mu\nu}v_{\rho_1\cdots\rho_k}$ is a nonzero multiple of $\gamma(F')T$, with F' the residual part of F and $T = \not{n}_1\cdots\not{n}_k$. On $\Sigma = \ker\gamma(\ell_+)$ the operator $\gamma(\ell_+\wedge\ell_-) = \frac{1}{2}(\gamma(\ell_+)\gamma(\ell_-)-\gamma(\ell_-)\gamma(\ell_+))$ acts as 1, and $\gamma(F') = F(\ell_-,\ell_+)\gamma(\ell_+\wedge\ell_-)$, so

$$\delta\lambda = c\,F(\ell_+,\ell_-)\,x, \qquad x = T\epsilon \in \Sigma, \tag{5.17}$$

for a constant $c \neq 0$; components of ϵ outside $\widehat{K}$ are annihilated by T (Lemma 5.2(iii)). In particular $\delta\lambda \in K$, and the gaugino constraint is preserved. For a vector Y, $\delta A(Y) = a\, B(\epsilon, \gamma(Y)\lambda)$. It vanishes for $Y = n_i$, because $\not{n}_i\lambda = 0$, and for $Y = \ell_+$, because $\gamma(\ell_+)\lambda = 0$. For $Y = \ell_-$ it is $a\,\beta(x, \gamma(\ell_-)\lambda)$, since B pairs $\mathbf{\Delta_1}$ only with $\widehat{K}$ (Lemma 5.2(iv)). With $\lambda = \psi e$ and $x = \theta e$, ψ and θ Grassmann-odd, this is $ab\,\theta\psi \neq 0$, which gives (N3). The constraint $n_i^\mu F_{\mu\nu} = 0$ is preserved, since $n_i^\mu \delta F_{\mu\nu} = D_{n_i}\delta A_\nu - D_\nu \delta A(n_i) = a\,\bar\epsilon\gamma_\nu D_{n_i}\lambda = 0$, and so is $D_{n_i}\lambda = 0$, since $D_{n_i}F = 0$ by the Bianchi identity and $\delta A(n_i) = 0$.

The gaugino equation and the cubic terms. In the basis $\ell_\pm, n_i, m_i$, whose dual basis is $\ell_\mp, m_i, n_i$,

$$\gamma^\mu D_\mu\lambda = \gamma(\ell_-)D_{\ell_+}\lambda + \gamma(\ell_+)D_{\ell_-}\lambda + \textstyle\sum_i(\not{n}_i D_{n_i}\lambda + \not{m}_i D_{m_i}\lambda) = \gamma(\ell_-)D_{\ell_+}\lambda,$$

because $D_\mu\lambda$ takes values in Σ and $D_{n_i}\lambda = 0$; as $\gamma(\ell_-)$ is injective on Σ, the gaugino equation is $D_{\ell_+}\lambda = 0$. By Lemma 5.5 the part of $\delta(\gamma^\mu D_\mu\lambda^I)$ quadratic in λ is $ag\, f^{IJK}\gamma^\mu\lambda^K\, B(\epsilon, \gamma_\mu\lambda^J)$. In the same basis the frame terms are $\not{m}_i\lambda^K B(\epsilon, \not{n}_i\lambda^J) + \not{n}_i\lambda^K B(\epsilon, \not{m}_i\lambda^J) = 0$, and the residual terms are $\gamma(\ell_+)\lambda^K\, B(\epsilon, \gamma(\ell_-)\lambda^J) + \gamma(\ell_-)\lambda^K\, B(\epsilon, \gamma(\ell_+)\lambda^J) = 0$, since $\gamma(\ell_+)$ annihilates Σ. This proves (ii), for every ϵ and not only for $\epsilon \in \widehat{K}$.

Closure. By (N4), $\xi(Y) = c'\,\bar\epsilon_2\gamma(Y \wedge v)\epsilon_1$. For $Y = n_i$ it vanishes, and for $Y \in V'$ it is a multiple of $\beta(x_2, \gamma(Y)x_1)$, which vanishes for $Y = \ell_+$. Hence $\xi = \xi^+\ell_+ + \sum_i c_i n_i$ with ξ^+ a nonzero multiple of $b\,\theta_2\theta_1$; this is (iv), and it gives (N4), because the bracket $\beta(x_2, \gamma(\ell_-)x_1)$ is symmetric in the commuting spinor indices of the one-dimensional space Σ. Since $F(n_i, \cdot) = 0$ and $D_{n_i}\lambda = 0$, the translation by ξ acts as the translation by $\xi^+\ell_+$. On λ, (5.17) and $\delta F(\ell_+, \ell_-) = D_{\ell_+}\delta A(\ell_-) - D_{\ell_-}\delta A(\ell_+) = ab\,\theta\, D_{\ell_+}\psi$ give

$$[\delta_1, \delta_2]\lambda = c\,ab\,(\theta_1(D_{\ell_+}\psi)\theta_2 - \theta_2(D_{\ell_+}\psi)\theta_1)e = 2abc\,\theta_2\theta_1\, D_{\ell_+}\lambda,$$

the translation along ℓ_+, without use of any field equation. On A, the components $A(n_i)$ are invariant and $F(\xi, n_i) = 0$; the component along ℓ_+ is invariant and $F(\xi^+\ell_+, \ell_+) = 0$; and the component along ℓ_- changes by

$$[\delta_1, \delta_2]A(\ell_-) = a\big(\beta(x_2, \gamma(\ell_-)\delta_1\lambda) - \beta(x_1, \gamma(\ell_-)\delta_2\lambda)\big) = 2abc\,\theta_2\theta_1\, F(\ell_+, \ell_-),$$

which is $F(\xi^+\ell_+, \ell_-)$ with the same coefficient. The translation of a gauge field by ξ is $\xi^\nu F_{\nu\mu}$ up to a gauge transformation with parameter $-\xi^\nu A_\nu$, and the remaining components $A(m_i)$ change by the extra transformation $\delta_E A_\mu = v_{\mu\nu_1\cdots\nu_{k-1}}\Omega^{\nu_1\cdots\nu_{k-1}}$ of Nishino's eq. (2.10), which shifts each $A(m_i)$ independently. This proves (iii), and (i) follows from the preceding steps. For $k = 0$ the frame is empty, $T = 1$, and (5.16) is $\delta A_\mu = a\,\bar\epsilon\gamma_\mu\lambda$, $\delta\lambda = \frac{1}{4}\gamma^{\mu\nu}F_{\mu\nu}\epsilon$. □

Corollary 5.15. *Let $s - t \equiv 0 \pmod 8$ and let $\mathbb{R}^{s,t}$ carry a real null frame of rank k. A null-projected super Yang–Mills theory of rank k with a Majorana–Weyl gaugino exists if and only if either $s + t - 2k = 10$ and $s - t \in \{8, 0, -8\}$, or $s + t - 2k = 2$ and $s = t$. Thus the conditions of the Main Theorem are necessary and sufficient.*

Proof. Necessity is Theorem 5.8. Sufficiency is Theorem 5.12 for residual dimension ten and Theorem 5.14 for residual dimension two. □

The theories of Theorem 5.14 are chiral: the gaugino moves along ℓ_+ only, and supersymmetry translates along the same direction, so on shell the commutator acts trivially on λ. Their cubic gaugino terms vanish for the same reason as in four, six and ten dimensions: the residual spinors satisfy the identity (1.1), here trivially, because the only nonzero component of the vector bilinear is along ℓ_+ and $\gamma(\ell_+)$ annihilates them. None of the families (1.2) reaches residual dimension two, since there $s - t \geq 16$.

6. A single time: counting states

For the first family in (1.2) the time is single, and representation theory gives a bound that does not depend on how the theory is constructed. Consider a unitary, Poincaré-invariant quantum theory in $\mathbb{R}^{9+8n,1}$ with $n \geq 1$, whose supercharges Q_α form one Majorana–Weyl

spinor, so that there are $S = 2^{4+4n}$ real supercharges. By Proposition 2.1 the most general anticommutator is

$$\{Q_\alpha, Q_\beta\} = (\gamma^\mu \mathcal{C})_{\alpha\beta} P_\mu + \sum_{p \in \{5,9,\dots,5+4n\}} (\gamma^{\mu_1 \cdots \mu_p} \mathcal{C})_{\alpha\beta} Z_{\mu_1 \cdots \mu_p}, \tag{6.1}$$

with $\mathcal{C}$ the charge conjugation matrix and $Z_{[p]}$ tensor charges, in accordance with the general structure of supersymmetry algebras [38, 34]. Tensor charges of this kind are carried by extended objects [39]; in ten dimensions the five-form charge is carried by five-branes. We state their absence on the massless multiplet as hypothesis (H).

Theorem 6.1. *Assume (H) that the tensor charges vanish on a massless supermultiplet. Then the multiplet has at least $2^{S/4} = 2^{2^{2+4n}}$ states for each momentum, half of them bosonic. In particular the theory has no massless vector multiplet: such a multiplet would have $D - 2 = 8 + 8n$ bosonic states for each generator of the gauge algebra.*

Proof. On the multiplet, with momentum p, $p^2 = 0$, $p^0 > 0$, the anticommutator is $M_{\alpha\beta} = (\not{p}\mathcal{C})_{\alpha\beta}$, a real symmetric matrix, positive semidefinite because the Q_α are Hermitian and the norm is positive. Its rank is $S/2$: choosing a real null frame of rank one with $n_1 = p$, Lemma 5.2 shows that $\not{p}$ annihilates exactly the half K of each Weyl module. A supercharge Q in the kernel of M has $\{Q, Q\} = 0$, so $\|Q\psi\|^2 = 0$ and Q vanishes on every state ψ of the multiplet. The remaining $S/2 = 2^{3+4n}$ combinations, rescaled, satisfy $\{\hat{Q}_a, \hat{Q}_b\} = 2\delta_{ab}$ and generate a complex Clifford algebra on 2^{3+4n} generators, whose irreducible representations have dimension $2^{2^{2+4n}}$ and are split evenly by $(-1)^F$. The multiplet is a direct sum of such representations. □

For $n = 0$ the bound is $2^4 = 16$: the eight bosonic and eight fermionic states of ten-dimensional super Yang–Mills theory. For $n = 1$, $D = 18$, it is 2^{64}, against sixteen bosonic and 128 fermionic states of a vector and a Majorana–Weyl gaugino. Figure 8 shows the gap for $n \le 3$. The argument is the one that bounds the helicity of massless supermultiplets in four dimensions by one for sixteen supercharges and by two for thirty-two [40, 41]. To exclude arbitrary multiplicities of vector, spinor and scalar fields, a state count alone is insufficient. The following helicity argument supplies the needed bound.

Lemma 6.2 (Helicity span)**.** *Under the hypotheses of Theorem 6.1, suppose the massless multiplet has a finite helicity spectrum for a rotation J in a transverse two-plane. If $h_{\min}$ and $h_{\max}$ are its least and greatest helicities, then*

$$h_{\max} - h_{\min} \ge S/8.$$

In particular a multiplet containing only scalar, spinor and vector polarizations has $S \le 16$, independently of their multiplicities.

Proof. Choose the null momentum in a four-dimensional Lorentzian subspace and let J generate rotations in its transverse two-plane. A transverse spinor has J-weights $+\frac{1}{2}$ and $-\frac{1}{2}$ with equal multiplicities. In its occupation basis the first occupation fixes this weight, while the chirality condition fixes only the parity of the remaining occupations; there are equally many choices for the two weights. On the $S/2$ real active supercharges, complexification therefore gives $S/4$ operators of each weight. The positive anticommutator of Theorem 6.1, invariant under J, pairs the two weights nondegenerately. After normalization the operators may be written $a_i^\dagger, a_i$, $1 \le i \le S/4$, with

$$\{a_i, a_j^\dagger\} = \delta_{ij}, \qquad \{a_i, a_j\} = \{a_i^\dagger, a_j^\dagger\} = 0, \qquad [J, a_i^\dagger] = \tfrac{1}{2} a_i^\dagger.$$

Take a unit vector Ω of helicity $h_{\min}$. Each a_i lowers helicity by $\frac{1}{2}$, so $a_i \Omega = 0$. The canonical anticommutation relations give

$$\left\| a_1^\dagger \cdots a_{S/4}^\dagger \Omega \right\|^2 = 1.$$

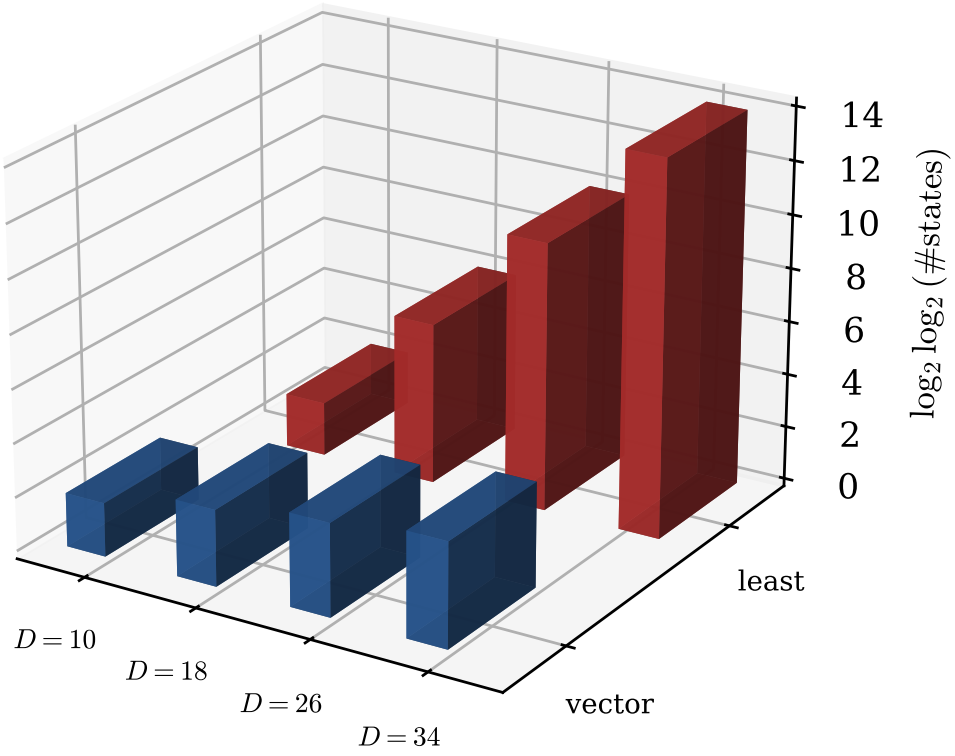


FIGURE 8. Massless multiplets in $(9{+}8n){+}1$, plotted as $\log_2 \log_2$ of the number of states. Blue: a vector field and a Majorana–Weyl gaugino, $(8 + 8n) + 2^{3+4n}$ states. Red: the least number $2^{2^{2+4n}}$ of states of a massless multiplet without tensor charges (Theorem 6.1). The two agree only for $D = 10$.

This nonzero vector has helicity $h_{\min} + S/8$, proving the span bound. A scalar has helicity 0, a spinor has helicities $\pm\frac{1}{2}$, and a vector has helicities $0, \pm 1$ under this rotation. Arbitrary multiplicities of these polarizations still have their spectrum in $[-1, 1]$. Thus $S/8 \leq 2$, or $S \leq 16$. □

For $n \geq 1$, $S = 2^{4+4n} > 16$, so a massless multiplet in this unitary, tensor-charge-free class cannot be assembled from any finite number of those low-spin fields. The mechanism is the familiar helicity bound on extended supersymmetry [40, 41]; it is written out here to distinguish it from the separate dimension count.

7. What this means for $27 + 3$ and $19 + 3$

In $27 + 3$ the Weyl spinor has $2^{14} = 16\,384$ components, and its symmetric square contains forms of degrees 3, 7, 11 and 15 only. The anticommutator of two supercharges contains no momentum. In the constructions of Refs. [19, 20, 22] translations are obtained by contracting a form charge with null vectors, as in (N4); the decomposition of the symmetric square does not by itself force this mechanism, and Sec. 7.2 treats the other contractions. With three times $k \leq 3$, so the residual space has dimension at least 24, and Corollary 5.9 applies: the cubic gaugino terms cannot be absorbed. The same holds in $19 + 3$, where the residual dimension is at least 16, and in every other signature of the families (1.2) with $n \geq 1$. For any compact non-abelian Lie gauge algebra, the obstruction (5.10) already arises on an $\mathfrak{su}(2)$

subalgebra and is unaffected by additional representation labels. A non-Lie extension or other field content would require assumptions beyond the present definition.

7.1. **The transfer of Nishino's construction.** The theories in $(11+8n)+3$ rest on one argument of Ref. [23, Sec. III.A]. Moving $4n$ timelike directions into space keeps $D=s+t$ and $s-t$ modulo eight, so the reality and symmetry properties of spinors and gamma matrices are unchanged, and the authors conclude that Nishino's existence proofs can be repeated in the new signatures.[8] The paper on $27+3$ takes this theory as given: its superalgebra (2.1) is (3.7) at $n=2$,

$$\{Q_\alpha,Q_\beta\}=(\gamma^{\mu\nu\rho})_{\alpha\beta}Z_{\mu\nu\rho}+(\gamma^{\mu_1\cdots\mu_7})_{\alpha\beta}Z_{\mu_1\cdots\mu_7}+(\gamma^{\mu_1\cdots\mu_{11}})_{\alpha\beta}Z_{\mu_1\cdots\mu_{11}}+(\gamma^{\mu_1\cdots\mu_{15}})_{\alpha\beta}Z_{\mu_1\cdots\mu_{15}}, \tag{7.1}$$

and it gives no transformation rules or field equations.[9] The quoted sentence names the proofs for (3.1) and (3.2), whose transfers are the first two families of (1.2); for those, Sec. 5 already shows that the repetition fails. The signatures $(11+8n)+3$ come from (3.3), the superalgebra of Nishino's Section 4, which treats $(11+4n)+(3+4n)$. That section is the proof to be repeated, and we write it out.

Nishino's coordinates in $(11+4n)+(3+4n)$ are $x^0,\dots,x^9,x^{11},\dots,x^{14+8n}$, with x^0 and $x^{12},x^{14},\dots,x^{14+8n}$ timelike [22, eq. (4.2)]. His null vectors, eqs. (4.5) and (4.6), are

$$n_{(i)}=\tfrac{1}{\sqrt{2}}(e_{9+2i}+e_{10+2i}),\qquad m_{(i)}=\tfrac{1}{\sqrt{2}}(e_{9+2i}-e_{10+2i}),\qquad i=1,\dots,4n+2, \tag{7.2}$$

each pairing one spacelike with one timelike unit vector, so the $4n+2$ vectors $n_{(i)}$ use $4n+2$ different times. With $v=n_{(1)}\wedge\cdots\wedge n_{(4n+2)}$ his theory consists of

$$\delta A^I_\mu=\bar\epsilon\gamma_\mu\lambda^I,\qquad \delta\lambda^I=-\tfrac14\gamma^{\mu\nu[4n+2]}\epsilon\,F^I_{\mu\nu}v_{[4n+2]}, \qquad \text{eq. (4.3)} \tag{7.3}$$

$$v^{[4n+1]}{}_\mu\gamma^\mu\lambda^I=0, \qquad \text{eq. (4.9)} \tag{7.4}$$

$$D_\mu F^{\mu[\nu_1 I}v^{\nu_2\cdots\nu_{4n+3}]}=\tfrac{1}{2^{4n+2}(4n+3)!}f^{IJK}\bar\lambda^J\gamma^{\nu_1\cdots\nu_{4n+3}}\lambda^K, \qquad \text{eq. (4.12)} \tag{7.5}$$

$$\{Q_\alpha,Q_\beta\}=(\gamma^{\mu[4n+2]})_{\alpha\beta}v_{[4n+2]}P_\mu, \qquad \text{eq. (4.15)} \tag{7.6}$$

together with the closure (4.13), (4.14), whose translation parameter is $\xi^\mu=-(\bar\epsilon_1\gamma^{\mu[4n+2]}\epsilon_2)v_{[4n+2]}$. Here $[p]$ denotes p antisymmetrized indices. The right-hand side of (7.5) is the source that absorbs the cubic gaugino terms, and (7.6) shows where the momentum comes from: the $(4n+3)$-form charge contracted with v. Repeating the proof in $(11+8n)+3$ means keeping these equations and choosing a null frame in the new signature. Since the rank $4n+2$ may be too large there, we allow any rank k.

Proposition 7.1. *Let $n\geq1$ and let $\mathfrak{g}$ be a compact non-abelian Lie algebra. In $\mathbb{R}^{11+8n,3}$ consider a gauge field A^I_μ, a Majorana–Weyl gaugino λ^I and a Majorana–Weyl parameter ϵ of the same chirality, as in (3.7) of Ref.* [23] *and* (7.1)*, a real null frame of rank $k\geq0$ with $v=n_1\wedge\cdots\wedge n_k$ ($v=1$ if $k=0$), a coupling $g\neq0$ in $D_\mu\lambda^I$, and Nishino's equations with $4n+2$ replaced by k, as identities for all gaugino values allowed by (b) and all ϵ:*

(a) $\delta A^I_\mu=\bar\epsilon\gamma_\mu\lambda^I$ and $\delta\lambda^I=-\frac14\gamma^{\mu\nu\rho_1\cdots\rho_k}\epsilon\,F^I_{\mu\nu}v_{\rho_1\cdots\rho_k}$, as in (7.3)*;*

[8] Ref. [23, pp. 046004-4 and 046004-5]: "The simple but crucial observation is that, by switching $4n$ timelike dimensions into $4n$ spacelike dimensions in the above space-time signatures, one keeps D and ρ unchanged, and so all the properties of spinors and γ-matrices relevant for the definition of the above super-Poincaré algebras are left unchanged. Thus, one can check that the proofs given in [6] (see also [5]) for the existence of SYM's with superalgebras (3.1) and (3.2) can be successfully repeated to show the existence of SYM's based on global minimal, chiral (1,0) superalgebras in the following space-time signatures". Here $\rho=s-t$ modulo eight, [6] is Ref. [22] and [5] is Ref. [20]; the list that follows is eqs. (3.5)–(3.8), and the third entry is $D=(11+8n)+3$ with the superalgebra (3.7).

[9] Ref. [24, p. 2]: "In $D=27+3$, the electric 11-brane has a multi-time worldvolume, with signature $11+3$, which can be used to provide a worldvolume realization for the $11+3$ SYM of Bars and Sezgin [7–9]." Its footnote 2 reads: "It should be recalled that the Bars-Sezgin SYM theories in $D>10$ are not Lorentz covariant, and so are the generalizations presented here." The lack of Lorentz covariance is the choice of the null vectors.

(b) $v_{\rho_1\cdots\rho_{k-1}\mu}\gamma^\mu\lambda^I = 0$ *if* $k \geq 1$, *as in* (7.4), *and the gaugino values are all Majorana spinors satisfying it;*
(c) *the commutator of two supersymmetries contains the translation by* $\xi^\mu = -(\bar\epsilon_1\gamma^{\mu\rho_1\cdots\rho_k}\epsilon_2)v_{\rho_1\cdots\rho_k}$, *the coefficient of* P_μ *in* $\{Q_\alpha, Q_\beta\} = (\gamma^{\mu\rho_1\cdots\rho_k})_{\alpha\beta}v_{\rho_1\cdots\rho_k}P_\mu$ *as in* (7.6), *and its residual components* $\xi^{\mu'}$ *do not vanish identically;*
(d) *the terms quadratic in* λ *in* $\delta(\gamma^\mu D_\mu\lambda^I)$ *are cancelled by a source* $J \wedge v$ *in the field equation of the gauge field, as in* (7.5).

Then:

(i) $k \leq 3$, *so Nishino's rank* $4n+2$ *cannot be kept.*
(ii) *Condition (c) holds only for* $k = 2$.
(iii) *For* $k = 2$ *conditions (a)–(c) imply (N1)–(N4), condition (d) is (N5), the residual space is* $\mathbb{R}^{9+8n,1}$, *and (d) fails: absorption would force the identity* (3.2) *on the residual Weyl module, of dimension* $N = 2^{4+4n}$, *while Theorem 3.3 requires* $N = 2(8+8n) = 16+16n$.

Hence no field theory in $(11+8n)+3$, *in particular none in* $19+3$ *or* $27+3$, *satisfies (a)–(d) for any rank of the null frame.*

Proof. We write $D = 14+8n = 2r$, so $r = 7+4n$, and use the oscillator notation of Sec. 5: K is the joint kernel of the $\not{n}_i$ and $\widehat{K}$ the joint kernel of the $\not{n}_i$. The gaugino does not vanish identically, since otherwise $\delta A = 0$ and no field transforms.

Step 1: the rank is at most three. First, (b) says exactly that λ^I takes values in K. Indeed $v_{\rho_1\cdots\rho_{k-1}\mu}\gamma^\mu$ is, for each choice of $\rho_1,\dots,\rho_{k-1}$, a linear combination of $\not{n}_1,\dots,\not{n}_k$, so $\not{n}_i\lambda = 0$ for all i implies (b). Conversely, contract (b) with m_j^ρ in the slots $\rho_1,\dots,\rho_{k-1}$ for all $j \neq i$; since $n_l \cdot m_j = \delta_{lj}$, only the term with n_i in the last slot survives, and (b) becomes a nonzero multiple of $\not{n}_i\lambda^I = 0$. Next, the n_i are real, mutually orthogonal null vectors, linearly independent because $n_i \cdot m_j = \delta_{ij}$, so they span a real totally isotropic subspace of dimension k, and Lemma 2.2 gives $k \leq \min(11+8n, 3) = 3$. Lemma 2.3 gives the same bound for any frame, real or complex, once the values of λ are Majorana spinors. Nishino's rank is $4n+2 \geq 6$, which proves (i). The same count holds in the source signature, where $\min(11+4n, 3+4n) = 3+4n \geq 4n+2$ leaves room for his frame.

Step 2: only rank two gives a translation. By Proposition 2.1, the forms $\bar\epsilon_1\gamma^{\mu_1\cdots\mu_p}\epsilon_2$ on $\Delta_+ \times \Delta_+$ vanish for even p, are symmetric in ϵ_1, ϵ_2 for $p \equiv r \equiv 3 \pmod 4$, and antisymmetric for $p \equiv 1 \pmod 4$. For $27+3$ the degrees are

$$\text{symmetric: } 3,7,11,15, \qquad \text{antisymmetric: } 1,5,9,13, \qquad \text{zero: } 0,2,4,\dots,14,$$

and the symmetric ones are the charges of (7.1). The parameter ξ^μ of (c) is the component of $\epsilon_1 \otimes \epsilon_2$ in degree $k+1$. For $k = 1$ and $k = 3$ the degree is even and ξ vanishes identically. For $k = 0$ the degree is one, the form is antisymmetric, and the momentum is absent from the anticommutator; this is why, among the superalgebras (3.1)–(3.8) of Ref. [23], only (3.1) and (3.5) contain P_μ. For $k = 2$ the degree is three and the form is symmetric. With $k \leq 3$ this leaves $k = 2$, which proves (ii). Nishino's rank $4n+2$ would draw the momentum from the $(4n+3)$-form charge, symmetric as well, contracted with a null form of rank $4n+2$; in $27+3$ this is the eleven-form charge of Ref. [24], and his contraction needs ten mutually orthogonal real null vectors, whereas by Step 1 there are at most three.

Step 3: at rank two the axioms (N1)–(N4) hold. Let $k = 2$. Axiom (N1) holds because, by hypothesis, the gaugino values are all spinors allowed by (b), and by Step 1 these form the full corner K_χ; ϵ ranges over the full module, as in Nishino's construction. Axiom (N2) is (a), with $a = 1$ and $\Phi(F)\epsilon = -\frac{1}{4}\gamma^{\mu\nu\rho_1\rho_2}\epsilon F_{\mu\nu}v_{\rho_1\rho_2}$, which is linear in F and independent of λ, and $g \neq 0$ by hypothesis. For (N3) and (N4) we use Lemma 5.2. Since k is even and λ and ϵ have the same chirality, $T\epsilon$ ranges over $K_\chi =: \Sigma$ as ϵ ranges over $\widehat{K}_\chi$, and Σ is a Weyl module of the residual space $V' = \mathbb{R}^{9+8n,1}$, of dimension $2^{r-k-1} = 2^{4+4n}$. As r is odd, B pairs Δ_+ with Δ_-, so β of (5.11) pairs Σ nondegenerately with the opposite Weyl half $\Sigma_{\rm opp}$ of $\Delta_{\mathbf{1}}$, and $\gamma^{\mu'}$ maps Σ injectively into $\Sigma_{\rm opp}$. Therefore $\delta A^I_{\mu'} = \beta(T\epsilon, \gamma_{\mu'}\lambda^I)$ and $\xi^{\mu'} = \pm\beta(T\epsilon_2, \gamma^{\mu'}T\epsilon_1)$, as

in (5.12), do not vanish identically, and the pairing in $\xi^{\mu'}$ is symmetric because $r' = r - 2 \equiv 1 \pmod 4$. Condition (d) asserts that the quadratic terms have the form (5.6), which is (N5).

Step 4: absorption fails. Theorem 5.6 now applies and gives $D' \in \{2, 10\}$, whereas $D' = 14 + 8n - 4 = 10 + 8n$. Concretely, Steps 1–3 of its proof turn (d) into the identity (3.2) on Σ with $d = D' = 10 + 8n$ and $N = 2^{4+4n}$. The second contraction (3.5) fixes the constant, $j = -(d-2)/N = -(8+8n)/2^{4+4n}$, and the first contraction (3.4) then requires $N = 2(d-2)$, which is false for every $n \geq 1$ because $2^{4+4n} = 16^{n+1} > 16(n+1)$ for $n \geq 1$. This proves (iii). □

For $27 + 3$ the numbers are $D' = 26$, $N = 4096$ and $j = -24/4096 = -3/512$, and (3.4) would require $4096 - 24 = -2j \cdot 24$, that is $4072 = 9/32$. In Nishino's own signatures the same count gives $N = 16 = 2(10-2)$, because his frame has rank $t-1$ and leaves the residual space $\mathbb{R}^{9,1}$. Compare Sezgin's theory: in $11+3$ a rank-two frame leaves ten residual dimensions and everything closes; in $27+3$ the same rank, the only one the translation bracket allows, leaves twenty-six. Table 1 and Figs. 9 and 10 collect the two cases singled out in Refs. [23, 24], and Appendix D checks each step in explicit gamma matrices of size 2^{15}.

target	source	Nishino's k	largest real k	k allowed by (c)	D'	N	$2(D'-2)$
$(19,3)$	$(15,7)$	6	3	2	18	256	32
$(27,3)$	$(19,11)$	10	3	2	26	4 096	48

TABLE 1. The transfer of Nishino's construction from $(11+4n)+(3+4n)$ to $(11+8n)+3$ for $n = 1, 2$ (Proposition 7.1). The rank of Nishino's frame exceeds the Witt index 3 of the target; the only rank compatible with the translation bracket is 2; and at rank 2 the residual Weyl module has N components where absorption needs $2(D'-2)$.

Remark 7.2. The transfer becomes consistent only by giving up the reality of the gaugino. The complexified Clifford algebra of $\mathbb{R}^{s,t}$ depends only on D, so Nishino's equations in $19+11$ can be written with a complex null frame of rank 10 in $\mathbb{C}^{30}$ and read in the coordinates of $\mathbb{R}^{27,3}$. The residual space is then ten-dimensional and the absorption holds, as in Remark 5.10. But the frame spans a complex totally isotropic subspace of dimension $10 > 3$, so by Lemma 2.3 its occupied corner contains no nonzero Majorana spinor: the gaugino, and through $\delta A = \bar{\epsilon}\gamma\lambda$ the gauge field, are complex. These are the complexified equations of the theory in $19+11$, whose real form there carries a real frame. The data that Ref. [23] keeps fixed, D and $s-t$ modulo eight, fix the complex Clifford algebra and the reality type of its spinors, but not the Witt index $\min(s,t)$ that a real null frame needs.[10]

7.2. **Other contractions.** Proposition 7.1 keeps the null contraction of Nishino's translation bracket. The superalgebra (7.1) admits others. Contracting its eleven-form charge with ten orthonormal spacelike vectors $e_1, \ldots, e_{10}$, for instance, gives $\{Q_\alpha, Q_\beta\} = (\gamma^{a\,1\cdots10})_{\alpha\beta} P_a$ with a orthogonal to the e_i and the P_a central, a consistent superalgebra with symmetry $\mathrm{Spin}(17,3) \times \mathrm{Spin}(10)$. The next theorem shows that no choice of this kind rescues the field theory. Whatever fixed tensor contracts the source in the field equation of the gauge field, the cubic gaugino terms cannot be absorbed once the residual dimension is sixteen or more. The theorem does not use (N4), and it includes a gaugino without any constraint ($k = 0$).

For a multivector $\omega \in \Lambda V_{\mathbb{C}}$ let $\gamma(\omega)$ be the antisymmetrized product of gamma matrices, as in Step 1 of the proof of Theorem 5.6. For a null frame, $V_{\mathbb{C}}$ is the direct sum of $\mathrm{span}(n_i)$, $\mathrm{span}(m_i)$ and $V'_{\mathbb{C}}$, and for $\omega \in \Lambda V_{\mathbb{C}}$ we write $\omega^\circ \in \Lambda V'_{\mathbb{C}}$ for the coefficient of $n_1 \wedge \cdots \wedge n_k$ in the part of ω that contains no m_i.

[10] The annihilator of a spinor is a totally isotropic subspace; spinors whose annihilator has the largest possible dimension are Cartan's pure spinors [32, 33]. Lemma 2.3 adds that for a Majorana spinor this subspace is the complexification of a real one.

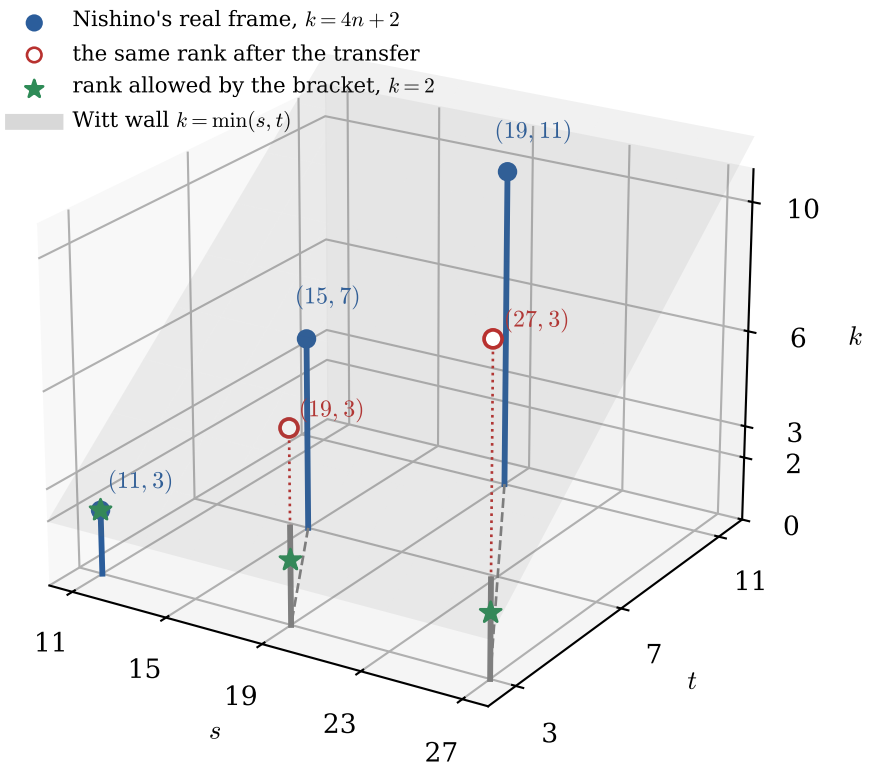


FIGURE 9. The transfer in the space of signatures and ranks (s, t, k). Blue: Nishino's real frames of rank $4n + 2$ in $(11 + 4n) + (3 + 4n)$ for $n = 0, 1, 2$, all below the wall $k = \min(s, t)$ (grey). Dashed: moving $4n$ times into space. Red circles: the same ranks in $(11 + 8n) + 3$, above the wall, since a real frame there has at most three null vectors. Green stars: the rank $k = 2$, the only one compatible with the translation bracket; at $n = 0$ it is Sezgin's theory in $11 + 3$.

Theorem 7.3. *Let $D = 2r$, let a null frame of rank $k \geq 0$, real or complex, be given (for $k = 0$ there is no frame and $K = \widehat{K} = \Delta$), and let $D' = D - 2k = 2r' \geq 2$. Suppose that the gaugino takes all values in K_χ, that (N2) and (N3) hold, and that there is a fixed $u \in \Lambda^q V_{\mathbb{C}}$ such that for every λ there are J^I, independent of ϵ, with*

$$(\text{part of } \delta(\gamma^\mu D_\mu \lambda^I) \text{ quadratic in } \lambda) = \gamma(J^I \wedge u)\,\epsilon \qquad \text{for every } \epsilon \in \widehat{K}_{\chi'}. \tag{7.7}$$

Let $P' = \{p : 0 \leq p < r',\ p \equiv r' \pmod 4\}$. Then P' has at most one element p, and $\binom{D'}{p} \leq D' + k$. Consequently $D' \leq 14$, and $D' \leq 10$ when $k \leq 53$.

Proof. Projection. As in Step 1 of the proof of Theorem 5.6, for $\epsilon \in \widehat{K}_{\chi'}$ the left-hand side of (7.7) is $ag\, f^{IJK} \gamma^{\mu'} \lambda^K\, \beta(x, \gamma_{\mu'} \lambda^J)$ with $x = T\epsilon$ and β as in (5.11), and it lies in $K = \Delta_{\mathbf{1}}$. On the right, expand $J \wedge u$ in the monomials $n_I \wedge m_L \wedge \omega'$ with $\omega' \in \Lambda V'_{\mathbb{C}}$. The factors for different i and the residual factor span mutually orthogonal subspaces, so γ of such a monomial is, up to sign, the product of $\gamma(\omega')$, of $\not{n}_i$ for $i \in I \setminus L$, of $\not{m}_i$ for $i \in L \setminus I$, and of $\gamma(n_i \wedge m_i) = P_i^{\downarrow} - P_i^{\uparrow}$ for $i \in I \cap L$. By Lemma 5.2(ii) it maps Δ_a into $\Delta_{a + e_{I \setminus L} - e_{L \setminus I}}$, so it maps $\widehat{K} = \Delta_{\mathbf{0}}$ into K only if $I = \{1, \dots, k\}$ and $L = \emptyset$. Projecting both sides onto K therefore gives, for every $x \in K_{\chi''}$,

$$ag\, f^{IJK} \gamma^{\mu'} \lambda^K\, \beta(x, \gamma_{\mu'} \lambda^J) = \pm \gamma((J^I \wedge u)^\circ)\, x. \tag{7.8}$$

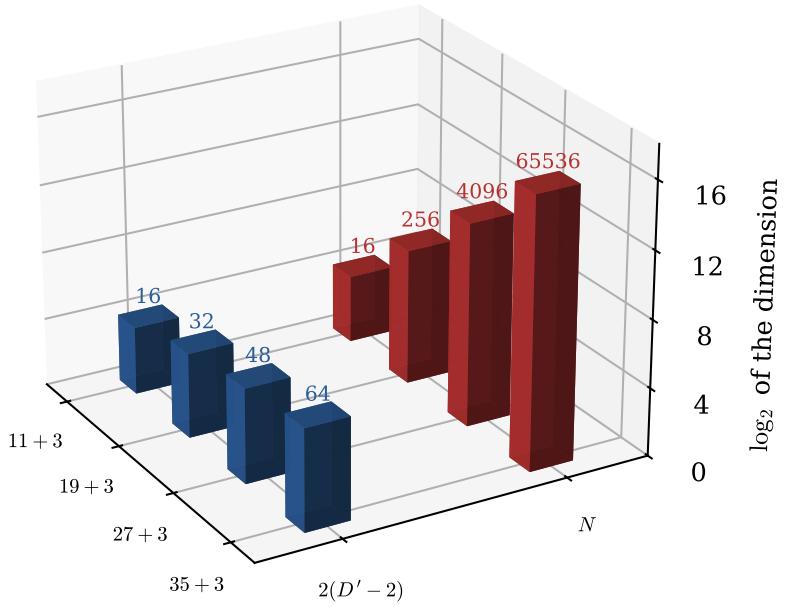


FIGURE 10. At rank two in $(11+8n)+3$, the dimension $N = 2^{4+4n}$ of the residual Weyl module (red) against the dimension $2(D'-2) = 16+16n$ that absorption of the cubic terms requires by Theorem 3.3 (blue), on a logarithmic scale. They agree only for $11+3$, Sezgin's theory.

Write $J = J' + \sum_i (a_i n_i + b_i m_i)$ with $J' \in V'_{\mathbb{C}}$. A term $b_i m_i$ produces an m_i, and $a_i n_i$ contributes only through the coefficient $u^{(i)} \in \Lambda V'_{\mathbb{C}}$ of $n_1 \wedge \cdots \wedge \widehat{n_i} \wedge \cdots \wedge n_k$ in the part of u without m's. Hence $(J \wedge u)^\circ = J' \wedge u^\circ \pm \sum_i a_i u^{(i)}$ lies in a subspace $E \subset \Lambda^{p_0} V'_{\mathbb{C}}$, $p_0 = q-k+1$, of dimension at most $D'+k$.

Degrees. By Lemma 5.2(i), K is a Dirac module of $\mathrm{Cl}(V'_{\mathbb{C}})$, and β is a nondegenerate form on it with $\beta(\gamma^{\mu'} x, y) = \pm\beta(x, \gamma^{\mu'} y)$, by Lemma 5.2(iii) and (2.1). Such a form is unique up to a factor, so it is symmetric or antisymmetric, and the proof of Proposition 2.1 applies to it: with $\Sigma = K_\chi$, the map that sends $S \in \mathrm{Sym}^2 \Sigma$ to its components $Y_p(S)$ in the forms of degree $p \equiv r' \pmod 4$, $p \le r'$, is an isomorphism onto $\bigoplus_p \Lambda^p V'_{\mathbb{C}}$. The left-hand side of (7.8) is $\pm ag \sum_{\mu'} \gamma^{\mu'} X^I \gamma_{\mu'} x$ with $X^I = f^{IJK} \lambda^K \beta(\lambda^J, \cdot\,)$, and in an orthonormal basis

$$\sum_{\mu'} \gamma^{\mu'} \gamma(\omega) \gamma_{\mu'} = (-1)^p (D' - 2p)\, \gamma(\omega) \qquad (\omega \in \Lambda^p V'_{\mathbb{C}}),$$

because $\gamma_{\mu'}$ anticommutes with the factors of a basis monomial that differ from it and commutes with the one equal to it. On $K_{\chi''}$ the left-hand side is therefore $\pm ag \sum_p (-1)^p (D' - 2p)\, \gamma(Y_p(S^I))$, summed over $p \equiv r' \pmod 4$, $p \le r'$, where the term with $p = r'$ vanishes. The right-hand side is $\gamma(e)$ with $e \in E \subset \Lambda^{p_0} V'_{\mathbb{C}}$, and on a Weyl module of V' the operators $\gamma(\Lambda^{p_0})$ and $\gamma(\Lambda^{D'-p_0})$ coincide by Hodge duality, while the operators $\gamma(\Lambda^p)$ with $p < r'$ of different degrees are independent. As in Step 3 of the proof of Theorem 5.6, the tensors S^I span $\mathrm{Sym}^2 \Sigma$, so for every $p \in P'$ the components $Y_p(S^I)$ fill $\Lambda^p V'_{\mathbb{C}}$, while $D' - 2p \neq 0$. Hence every $p \in P'$ equals $\min(p_0, D' - p_0)$, and then $\Lambda^p V'_{\mathbb{C}}$ is the image of E under Hodge duality or E itself, so $\binom{D'}{p} \le \dim E \le D' + k$.

For $r' \ge 8$ both $r'-4$ and $r'-8$ lie in P', so $D' \le 14$. For $D' = 12$ and $D' = 14$ one has $P' = \{2\}$ and $P' = \{3\}$, and $\binom{12}{2} = 66$, $\binom{14}{3} = 364$ exceed $D' + k$ unless $k \ge 54$. □

Corollary 7.4. *Let (s,t) be one of the signatures (1.2) with $n \geq 1$. Let the Majorana–Weyl gaugino take all values in the joint kernel of $k \geq 0$ mutually orthogonal null vectors, and let (N2) and (N3) hold. Then for no multivector u can the cubic gaugino terms be absorbed by a source of the form $J \wedge u$. In $27+3$ this covers every $k \leq 3$, the spacelike contraction above, every contraction with a mixture of null and non-null vectors, and the gaugino without constraint.*

Proof. By Lemma 2.3 the null vectors lie in a real totally isotropic subspace, so $k \leq \min(s,t) = t$, and they extend to a real null frame of rank k. Its residual dimension is $D' = s + t - 2k \geq s - t = 8 + 8n \geq 16$, which Theorem 7.3 excludes. □

Theorem 7.3 is weaker than Theorem 5.6 in one respect and stronger in two. Without (N4) it does not force $D' = 10$: for $D' \leq 6$ the projected cubic terms vanish identically, and for $D' = 8$ a source of degree seven in the residual directions absorbs them. But it allows any fixed tensor u, and its bound $D' \leq 14$ holds for every frame. Figure 11 shows the degrees that decide it. Appendix E checks the theorem in explicit gamma matrices.

7.3. **Several gauginos.** With several gauginos the natural extension of (N2) lets each gaugino transform into the field strength with its own parameter. This is the structure of extended supersymmetry, where a pairing of the gaugino and parameter labels that is invariant under the R-symmetry makes both transformation rules diagonal. A source in the field equation of the gauge field enters the variation of the field equation of λ_A only through $\delta\lambda_A$, so the cubic terms it can absorb are linear in ϵ_A; the cubic terms produced by the other parameters must vanish. The next proposition shows that they do not vanish once the residual dimension is at least three.

Proposition 7.5. *Let a null frame of rank $k \geq 0$, real or complex, have residual dimension $D' \geq 3$, and let $\mathfrak{g}$ be non-abelian. Consider $M \geq 2$ gauginos $\lambda_1, \dots, \lambda_M$ in the adjoint representation, where λ_A takes all values in K_{χ_A} for a chirality χ_A, with independent parameters $\epsilon_1, \dots, \epsilon_M$ and transformations*

$$\delta A^I_\mu = \sum_{A=1}^{M} a_A\, \bar\epsilon_A \gamma_\mu \lambda^I_A, \qquad \delta\lambda^I_A = \Phi_A(F^I)\,\epsilon_A, \tag{7.9}$$

where the linear maps Φ_A do not depend on λ and $D_\mu\lambda^I_A = \partial_\mu\lambda^I_A + g f^{IJK} A^J_\mu \lambda^K_A$ with $g \neq 0$. Suppose that the residual components $\delta A^I_{\mu'}$ do not vanish identically. Then for some A the part of $\delta(\gamma^\mu D_\mu \lambda^I_A)$ quadratic in λ does not vanish when $\epsilon_A = 0$, so it cannot be absorbed by a source.

Proof. For $\epsilon \in \widehat{K}_{\chi'}$ and $y \in K$ one has $B(\epsilon, \gamma_{\mu'} y) = \beta(T\epsilon, \gamma_{\mu'} y)$ by (5.11), and $B(\epsilon, \gamma_{\mu'} y) = 0$ for ϵ in the other corners, by Lemma 5.2(i) and (iv). Hence the residual components of δA are $\sum_A a_A \beta(T\epsilon_A, \gamma_{\mu'}\lambda_A)$, and by hypothesis there is an index B with $a_B \neq 0$ for which the residual vector $G(x,y)$ with components $G^{\mu'}(x,y) = \beta(x, \gamma^{\mu'} y)$ is not identically zero for $x \in K_{\chi''}$, $y \in \Sigma_B = K_{\chi_B}$.

Choose $A \neq B$, which is possible as $M \geq 2$, and put $\epsilon_C = 0$ for $C \neq B$. Then $\delta\lambda_A = 0$, and the part of $\delta(\gamma^\mu D_\mu \lambda^I_A)$ quadratic in λ is $a_B g\, f^{IJK} \gamma^\mu \lambda^K_A\, (\bar\epsilon_B \gamma_\mu \lambda^J_B)$. In the basis n_i, m_i and a basis of V' the frame terms are $\not{m}_i \lambda_A\, B(\epsilon_B, \not{n}_i \lambda_B) + \not{n}_i \lambda_A\, B(\epsilon_B, \not{m}_i \lambda_B) = 0$, since $\not{m}_i$ annihilates K, and what remains is

$$a_B g\, f^{IJK}\, \gamma(G(x, \lambda^J_B))\lambda^K_A, \qquad x = T\epsilon_B,$$

with $\gamma(g) = g_{\mu'}\gamma^{\mu'}$. Take $\lambda^J_B = y\,\psi$ for a single value of J and $\lambda_B = 0$ otherwise, $\lambda^K_A = \sigma\,\varphi$ for a single K with $f^{IJK} \neq 0$ for some I, and $x = \tilde x\,\vartheta$, where $\tilde x$, y, σ are commuting spinors and ϑ, ψ, φ are Grassmann-odd. Since λ_A and λ_B are independent fields, the product $\vartheta\psi\varphi$ is not zero, so if the cubic terms vanished for $\epsilon_A = 0$ we would have

$$\gamma(G(\tilde x, y))\sigma = 0 \qquad \text{for all } \tilde x \in K_{\chi''},\ y \in \Sigma_B,\ \sigma \in \Sigma_A.$$

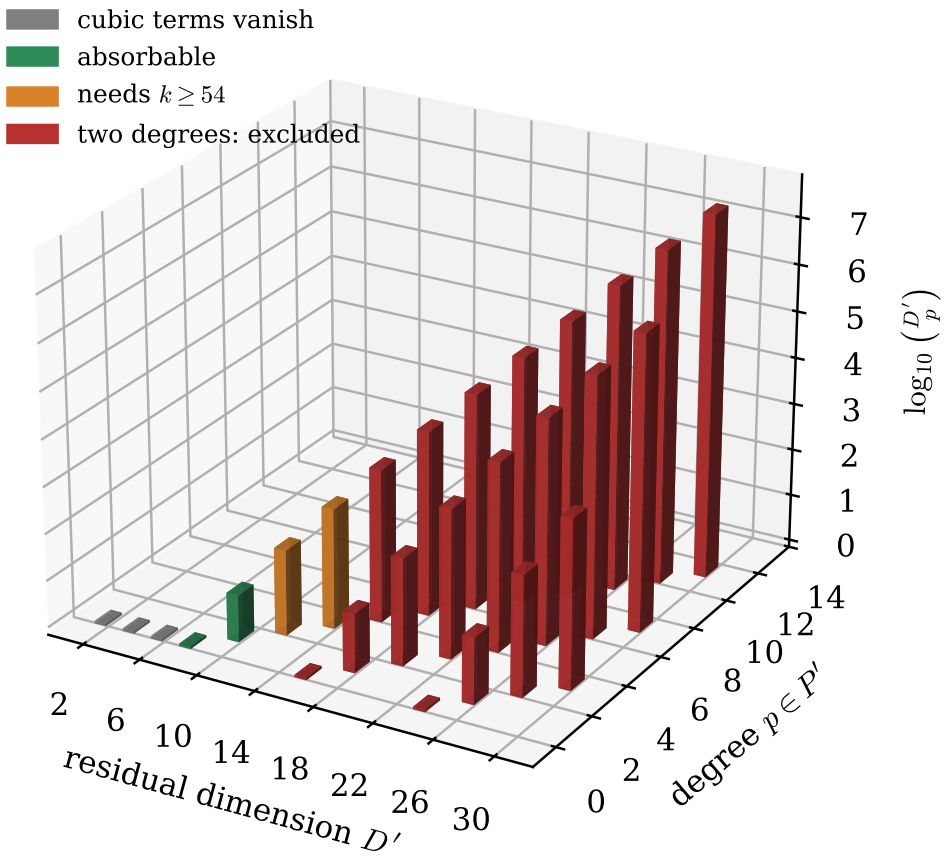


FIGURE 11. The degrees $p \in P'$ in which the projected cubic terms have nonzero components (Theorem 7.3), for residual dimensions $D' = 2, \ldots, 30$. Bar heights are $\log_{10}\binom{D'}{p}$, the dimension of the forms that a source must supply in degree p. Grey: P' is empty and the cubic terms vanish after projection. Green: one degree of dimension at most D', so a source can absorb them ($D' = 8$ and $D' = 10$). Orange: one degree, too large unless the frame has rank at least 54 ($D' = 12$, 14). Red: two or more degrees, which no source of a single degree can supply ($D' \geq 16$, where every signature of (1.2) with $n \geq 1$ lies).

Let $W = \{g \in V'_{\mathbb{C}} : \gamma(g)\Sigma_A = 0\}$. It contains a nonzero vector $G(\tilde{x}, y)$. As Σ_A is invariant under $\mathrm{Spin}(V')$ and $s\gamma(g)s^{-1} = \gamma(s \cdot g)$, the subspace W is invariant under $\mathrm{Spin}(V')$, hence under its Lie algebra and, being complex, under $\mathfrak{so}(V'_{\mathbb{C}})$. For $D' \geq 3$ the vector representation of $\mathfrak{so}(V'_{\mathbb{C}})$ is irreducible, so $W = V'_{\mathbb{C}}$. But $V'_{\mathbb{C}}$ contains a vector g with $\eta(g, g) \neq 0$, and then $\gamma(g)^2 = \eta(g, g)$ makes $\gamma(g)$ invertible, so $\gamma(g)\Sigma_A \neq 0$. This contradiction proves the proposition. □

The proof uses neither (N4) nor the form of the source, and it holds for every frame, real or complex, and for gauginos of either chirality. For $D' = 2$ it gives nothing: there $G(x, y)$ is a multiple of ℓ_+, which annihilates Σ, and the cross terms vanish, as in the proof of Theorem 5.14. Appendix F checks in explicit gamma matrices that the cross terms are nonzero for $D' = 4, 6, 8, 10$ and vanish for $D' = 2$.

Remark 7.6. The results of this section concern theories whose fields and transformation rules are written down, and no statement about the superalgebra alone can exclude every theory in $27 + 3$. The superalgebra (7.1) acts on functions of Grassmann coordinates θ^α and

of commuting coordinates $y^{\mu_1\cdots\mu_p}$, antisymmetric in their indices, for $p = 3, 7, 11, 15$: the operators

$$Q_\alpha = \frac{\partial}{\partial\theta^\alpha} + \frac{1}{2}\sum_p \sum_{\mu_1<\cdots<\mu_p} (\gamma^{\mu_1\cdots\mu_p}\theta)_\alpha \frac{\partial}{\partial y^{\mu_1\cdots\mu_p}}$$

satisfy (7.1) with $Z_{\mu_1\cdots\mu_p} = \partial/\partial y^{\mu_1\cdots\mu_p}$, because the matrices $\gamma_{\alpha\beta}^{\mu_1\cdots\mu_p}$ of these degrees are symmetric. Such a realization carries no dynamics, and nothing in this paper bears on it. What is excluded is a field content with specified transformation rules: one constrained gaugino with the null contraction of the translation bracket (Theorem 5.8), one gaugino, constrained or not, with any contraction of the source (Theorem 7.3), and several gauginos that transform with their own parameters (Proposition 7.5). Theories with auxiliary fields, with transformation rules nonlinear in the gaugino, with gauginos mixed by their transformation rules, with a gauge structure that is not a Lie algebra, or with a mechanism carried by the 11-brane lie outside these results.

7.4. **Scope.** The results leave the algebraic structures of Refs. [23, 27, 26] untouched. The superalgebras (3.5)–(3.8) of Ref. [23] exist by Proposition 2.1, and so do the cosets and Jordan-algebraic constructions built on them. The obstruction concerns realization by the specific null-projected field theory defined here. Applications invoking a $27 + 3$ theory by this particular transfer, including a worldvolume theory for M-theory [24] and the dark matter model of Ref. [25], require a construction outside these five assumptions, or a change to the claimed transfer; by Theorem 7.3, changing only the contraction of the source does not suffice. Every theory satisfying (N1)–(N5) obeys the necessary condition $s - t \in \{8, 0, -8\}$. For residual dimension greater than two its cubic fermion structure satisfies the ten-dimensional identity, and by Theorem 5.12 a theory exists in every signature with residual dimension ten that the Main Theorem leaves. For residual dimension two, which the Main Theorem allows when $s = t$, Theorem 5.14 gives a theory in every such signature, with vanishing cubic terms. This agrees with the view, expressed already in Ref. [21], that the null projections of super Yang–Mills theories beyond ten dimensions encode ten-dimensional theories with a covariant presentation of the extra directions.

Two remarks fix the scope of the results. The Main Theorem concerns theories with a single gaugino of definite chirality and the transformation rules (N2). Proposition 7.5 excludes several gauginos that transform with their own parameters, for every residual dimension of at least three. Theories with new auxiliary fields, with transformation rules nonlinear in λ, or with gauginos mixed by their transformation rules lie outside all of these results (Remark 7.6). Theorem 6.1 assumes unitarity and hypothesis (H), which is why it covers only the single-time family: we do not assume the positive-energy unitary representation used in the counting argument for the several-time signatures. The algebraic Theorem 5.8 is the result applied there.

Remark 7.7. The axioms (N1)–(N5) are conditions on transformation rules and field equations, and they can be checked only for a theory whose equations are written down. By Corollary 5.9 no theory in $27 + 3$ satisfies them, by Corollary 7.4 none satisfies (N1)–(N3) and absorbs the cubic terms with any other contraction of the source, and by Proposition 7.5 adding gauginos that transform with their own parameters does not help. The content of Proposition 7.1 is that the one field-theoretic construction proposed for $27 + 3$, the repetition of Nishino's proof in Ref. [23, Sec. III.A], would satisfy them, and therefore does not exist. The superalgebra (2.1) of Ref. [24] lists the possible central charges, but it contains no gaugino constraint, transformation rule or field equation, so by itself it can neither satisfy nor violate the axioms. A theory in $27 + 3$ that realizes it by other means, with several gauginos mixed by their transformation rules, auxiliary fields, transformation rules nonlinear in λ, a gaugino constraint other than annihilation by null vectors, a gauge structure that is not a Lie algebra, such as the Magic Star algebras of Ref. [27], which are not Lie algebras, or a mechanism carried by the 11-brane, is not decided here. The worldvolume theory invoked in Ref. [24]

is the theory of Sezgin in $11+3$ [20], Nishino's case $n=0$ with a frame of rank two. It satisfies (N1)–(N5) with residual space $\mathbb{R}^{9,1}$ and $s-t=8$, in agreement with Theorem 5.8, and nothing in this paper bears on whether the 11-brane in $27+3$ carries it.

8. Conclusions

Supersymmetric Yang–Mills theories built with null projections in signatures with several times are governed by an invariant that the Clifford algebra does not see. The bound $\min(s,t)$ limits the number of null directions that can be projected out of a real gaugino, and a residual dimension greater than two must equal ten because the absorption equation forces the ten-dimensional Fierz identity. The dimension count $N=2(d-2)$ of Theorem 3.3 makes the second statement rigid, and together the two leave only the signatures with $s-t\in\{8,0,-8\}$. The families $(9+8n)+1$, $(10+8n)+2$, $(11+8n)+3$ and $(12+8n)+4$ with $n\geq 1$, including $27+3$ and $19+3$, are excluded; in particular the repetition of Nishino's proof proposed for these signatures fails for every rank of the null frame (Proposition 7.1), and no other contraction of the source in the field equation of the gauge field repairs it (Theorem 7.3). The conditions are also sufficient. For residual dimension ten Nishino's theories carry over to every signature with $s-t\in\{8,0,-8\}$ and a real frame of the same rank, among them $9+9$ and $9+17$ (Theorem 5.12), and for residual dimension two his transformations give theories in every signature $(k+1)+(k+1)$, with vanishing cubic terms and off-shell closure (Theorem 5.14). Several gauginos that transform with their own parameters are impossible once the residual dimension is three or more (Proposition 7.5). For $(9+8n)+1$, Theorem 6.1 and Lemma 6.2 exclude massless vector multiplets and finite multiplets made only of scalar, spinor and vector fields under the stated unitarity and tensor-charge hypotheses. A supersymmetric gauge theory in these signatures would need auxiliary fields, transformation rules nonlinear in the gaugino, gauginos mixed by their transformation rules or a gauge structure that is not a Lie algebra, and it would then no longer be a repetition of Nishino's proof.

Data availability

The general results are proved in the text; numerical samples are supplementary. The plotted and tabulated residuals of Appendix B retain the archived reference run of the 34 checks of `verify.py`, which a fresh run on 24 October 2026 reproduced. The 24 checks of Appendix D are made by `transfer_check.py`, the 32 checks of Appendix E by `contraction_check.py`, and the 50 checks of Appendix F by `residual_two_check.py`; all four programs build their gamma matrices with `clifford.py`. Figures 7 and 11 are drawn by `make_v4_figures.py`. The complete LaTeX source, the four numerical programs and their verification logs are archived with every earlier version of this manuscript under one concept record [42]. The schematics of Figs. 2 and 5 are drawn in the LaTeX source with TikZ.

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Appendix A. The oscillator decomposition

We prove Lemma 5.2. The $P_i^{\uparrow}$ are commuting projections, so the products $\Pi_a=\prod_i(a_iP_i^{\uparrow}+(1-a_i)P_i^{\downarrow})$ are orthogonal idempotents with $\sum_a\Pi_a=1$, and $\Delta=\bigoplus_a\Delta_a$ with $\Delta_a=\Pi_a\Delta$. For $j\neq i$, $\not{n}_i$ anticommutes with $\not{n}_j$ and $\not{m}_j$ and therefore commutes with $P_j^{\uparrow}$. From $\not{n}_i^2=0$ and $\not{n}_i\not{m}_i+\not{m}_i\not{n}_i=2$ we get $\not{n}_i\not{m}_i\not{n}_i=2\not{n}_i$, that is $P_i^{\uparrow}\not{n}_i=\not{n}_i$ and $\not{n}_iP_i^{\uparrow}=0$. Hence $\not{n}_i$ annihilates Δ_a when $a_i=1$ and maps Δ_a into Δ_{a+e_i} when $a_i=0$; on the latter $\not{m}_i\not{n}_i=2P_i^{\downarrow}=2$, so the

map is injective. The same argument with $\not{n}_i$ gives an injection $\Delta_{a+e_i} \to \Delta_a$, so both maps are isomorphisms. This proves (ii), and all the Δ_a have the same dimension $2^r/2^k$.

Each $\gamma^{\mu'}$ anticommutes with every b_i and $b_i^\dagger$, hence commutes with every $P_i^\uparrow$ and preserves every Δ_a. Therefore Δ_a is a module over $\mathrm{Cl}(V'_{\mathbb{C}})$, which for even D' is a simple algebra of dimension 2^{2r-2k} with a unique irreducible module, of dimension 2^{r-k}. As $\dim \Delta_a = 2^{r-k}$, Δ_a is that module. This proves (i).

For (iii), up to the sign $(-1)^{k(k-1)/2}$, T is the composition of the isomorphisms of (ii) along the path $\mathbf{0} \to e_1 \to e_1 + e_2 \to \cdots \to \mathbf{1}$, it annihilates every Δ_a with $a \neq \mathbf{0}$ by (ii), and $\gamma^{\mu'}$ anticommutes with each of its factors. For (iv), by (2.1), $B(P_i^\uparrow \psi, \chi) = \frac{1}{2}\theta^2 B(\psi, \not{n}_i \not{n}_i \chi) = B(\psi, P_i^\downarrow \chi)$. So if $\psi \in \Delta_a$, $\chi \in \Delta_b$ and $a_i = b_i = 1$, then $B(\psi, \chi) = B(P_i^\uparrow \psi, \chi) = B(\psi, P_i^\downarrow \chi) = 0$, and similarly if $a_i = b_i = 0$. For (v), choose an orthonormal basis of $V_{\mathbb{C}}$ consisting of a basis of $V'_{\mathbb{C}}$ and the vectors $e_i = (n_i + m_i)/\sqrt{2}$, $f_i = (n_i - m_i)/\sqrt{2}$, with $e_i^2 = 1$ and $f_i^2 = -1$. Then $\not{e}_i \not{f}_i = \frac{1}{2}(\not{n}_i \not{m}_i - \not{m}_i \not{n}_i) = P_i^\downarrow - P_i^\uparrow$, and the chirality operator, proportional to the product of all basis gamma matrices, is $\Gamma_* = \pm \Gamma'_* \prod_i (P_i^\downarrow - P_i^\uparrow)$, with the sign fixed by $\Gamma_*^2 = 1$. On Δ_a the product equals $(-1)^{|a|}$.

Lemma A.1 (Null-frame current)**.** *Let $\lambda_1, \lambda_2 \in K$. After identifying vectors and covectors with the metric, the $(k+1)$-form*

$$\omega_{\mu_0 \cdots \mu_k} = B(\lambda_1, \gamma_{\mu_0 \cdots \mu_k} \lambda_2)$$

has the form $v \wedge j$ for a residual one-form $j \in (V')^*_{\mathbb{C}}$.

Proof. Evaluate ω on the null-frame basis n_i, m_i and a basis of V'. Both spinors are in the fully occupied corner. By Lemma 5.2(iv), a Clifford word pairing nontrivially with the first spinor must take the second to the empty corner. It must therefore contain every lowering operator $\not{n}_i$. A word of degree $k+1$ can do this only by using exactly these k operators and one residual generator: an additional raising operator would require another lowering operator and hence at least $k+2$ factors. Antisymmetrization preserves this occupancy count. Thus a nonzero component has one argument m_i for every i and one residual argument. Since $n_i \cdot m_j = \delta_{ij}$, these are exactly the components of $v \wedge j$. □

Appendix B. Numerical checks in explicit gamma matrices

The program `verify.py` in the repository builds the gamma matrices of $\mathbb{R}^{s,t}$ as Kronecker products of Pauli matrices, multiplying the first t Euclidean generators by i, and checks the Clifford relations. It finds a charge conjugation matrix $\mathcal{C}$ with $\mathcal{C}\gamma^\mu\mathcal{C}^{-1} = \pm(\gamma^\mu)^T$ among the products of the real or of the imaginary Euclidean generators, and builds the vector forms of Definition 3.1 from it. It then runs the following checks, all of which pass (34 checks, no failures; runtime depends on the numerical libraries and hardware).

(a) Proposition 2.1: the dimension count for $r = 2, \ldots, 20$, and the form degrees of the four families (1.2) for $n = 0, 1, 2$.
(b) The identity (1.1) on random spinors: it holds in $(9,1)$, $(5,5)$ and $(1,9)$ (relative residual at most 1.3×10^{-16}) and fails in $(17,1)$, $(13,5)$, $(9,9)$ and $(25,1)$ (relative residual between 0.09 and 0.36).
(c) The identity (3.2) with j fitted by least squares: $j = -0.5000$ in $(9,1)$ and $(5,5)$; no j fits in $(17,1)$ and $(25,1)$ (residual at least 0.99).
(d) Lemma 3.2(b),(c) in $(9,1)$ and $(17,1)$, exactly.
(e) The absorption equation (5.10), solved by least squares for J over random $\lambda \in K$ and $\epsilon \in \widehat{K}$ with explicit real frames $\not{n}_i = (\gamma_{\text{space}} + \gamma_{\text{time}})/\sqrt{2}$; see Table B.1 and Fig. 6.
(f) Lemma 2.3 in $(17,1)$ with a complex frame of rank four: the complex conjugate of each n_i is m_i, and $\not{n}_i \not{n}_i + \not{n}_i \not{n}_i = 2$, so the occupied corner contains no Majorana spinor.
(g) Lemma 2.2: real null frames of rank $\min(s,t)$ satisfying (5.1).

(h) Theorem 6.1: the counts of Fig. 8 for $n = 0, 1, 2, 3$.

(s,t)	k	frame	D'	$\dim K_\chi$	residual	solvable
$(9,1)$	0	–	10	16	3.0×10^{-15}	yes
$(10,2)$	1	real	10	16	1.6×10^{-15}	yes
$(11,3)$	2	real	10	16	1.5×10^{-15}	yes
$(12,4)$	3	real	10	16	1.4×10^{-15}	yes
$(13,5)$	4	real	10	16	2.6×10^{-15}	yes
$(6,6)$	1	real	10	16	1.4×10^{-15}	yes
$(7,7)$	2	real	10	16	2.8×10^{-15}	yes
$(17,1)$	0	–	18	256	0.98	no
$(18,2)$	1	real	18	256	0.99	no
$(19,3)$	0	–	22	1024	1.0	no
$(19,3)$	2	real	18	256	0.99	no
$(20,4)$	1	real	22	1024	1.0	no
$(20,4)$	3	real	18	256	0.98	no
$(17,1)$	4	complex	10	16	1.7×10^{-15}	yes (complex sector)
$(19,3)$	2	complex	18	256	0.97	no

TABLE B.1. The absorption equation (5.10) in explicit gamma matrices. The residual is the relative norm of the least-squares remainder over three random draws. These samples are consistent with the analytic obstruction of Theorem 5.6 for $D' > 2$; the complex frame in $(17,1)$ is the case of Remark 5.10.

APPENDIX C. SYMMETRIC SQUARES FOR THE FOUR FAMILIES

Table C.1 lists $\mathrm{Sym}^2 \Delta_+$, by Proposition 2.1, for the families (1.2) with $n = 0, 1, 2$, together with the residual dimension $s - t$ of a real null frame of the largest rank $k = t$. The momentum Λ^1 occurs only in the first family. Since D' decreases as k grows, a residual dimension of ten is reachable only when $s - t \le 10$, that is only for $n = 0$.

(s,t)	D	$\dim \Delta_+$	degrees in $\mathrm{Sym}^2 \Delta_+$	$\dim \mathrm{Sym}^2 \Delta_+$	D' at $k = t$
$(9,1)$	10	16	5, 1	136	8
$(10,2)$	12	32	6, 2	528	8
$(11,3)$	14	64	7, 3	2 080	8
$(12,4)$	16	128	8, 4, 0	8 256	8
$(17,1)$	18	256	9, 5, 1	32 896	16
$(18,2)$	20	512	10, 6, 2	131 328	16
$(19,3)$	22	1 024	11, 7, 3	524 800	16
$(20,4)$	24	2 048	12, 8, 4, 0	2 098 176	16
$(25,1)$	26	4 096	13, 9, 5, 1	8 390 656	24
$(26,2)$	28	8 192	14, 10, 6, 2	33 558 528	24
$(27,3)$	30	16 384	15, 11, 7, 3	134 225 920	24
$(28,4)$	32	32 768	16, 12, 8, 4, 0	536 887 296	24

TABLE C.1. Symmetric squares of the Weyl spinor for the families (1.2), $n = 0, 1, 2$. The last column is $s - t = s + t - 2\min(s,t)$, the residual dimension of a real null frame of rank $k = t$. For $n = 0$ Nishino's frames have $k = t - 1$ and $D' = 10$; a frame with $k = t$ gives $D' = 8$, which Theorem 5.6 excludes. For $n \ge 1$ every admissible frame has $D' \ge 16$.

Appendix D. The transfer in explicit gamma matrices

The program `transfer_check.py` in the repository checks Proposition 7.1 in the gamma matrices of Appendix B, which in thirty dimensions have size $2^{15} = 32\,768$. Real frames pair the i-th timelike generator with a spacelike one, $\not{n}_i = (\gamma_{\text{space}} + \gamma_{\text{time}})/\sqrt{2}$, as in Nishino's eqs. (4.5) and (4.6). Dimensions of corners are traces of the projections $\prod_i P_i^{\uparrow}$ restricted to a chirality, so they are exact integers. For the cubic terms we take $x = T\epsilon$ with a random $\epsilon \in \widehat{K_+}$, use the residual pairing $G^{\mu'}(T\epsilon, T\epsilon') = B(\epsilon, \gamma^{\mu'} T\epsilon')$ of Step 2 of the proof of Theorem 5.6, and evaluate $Y = \sum_{\mu'} \eta_{\mu'\mu'} G^{\mu'}(x,x)\gamma^{\mu'} x$, which vanishes for every x exactly when the identity (1.1) holds; the identity (3.2) is tested with j fitted by least squares. Sezgin's $11+3$ and Nishino's source signature $19+11$ serve as controls. All 24 checks pass; Table D.1 lists them. They confirm the written proof and do not replace it.

signature	check	result
all four	Clifford relations, spinor dimension $2^{D/2}$	hold
$(27,3)$, $(19,11)$	degrees symmetric on $\Delta_+ \times \Delta_+$	$3, 7, 11, 15$; antisymmetric $1, 5, 9, 13$; even degrees zero
$(19,3)$	degrees symmetric on $\Delta_+ \times \Delta_+$	$3, 7, 11$; antisymmetric $1, 5, 9$; even degrees zero
$(27,3)$, $(19,3)$	translation bracket for real frames of rank 1 and 3	vanishes identically
$(27,3)$, $k=3$	real frame, $\dim K_+$, D'	exists; 2 048; 24
$(27,3)$, $k=2$	real frame, $\dim K_+$, D'	exists; 4 096; 26
$(27,3)$, $k=2$	$\delta A_{\mu'}$, $\xi^{\mu'}$ nonzero, ξ symmetric	yes (asymmetry 8×10^{-16})
$(27,3)$, $k=2$	identity (1.1); identity (3.2) for some j	fail ($\lvert Y\rvert$ relative 0.11; residual 0.999)
$(19,3)$, $k=2$	$\dim K_+$, D'; $\delta A_{\mu'}$, $\xi^{\mu'}$	256, 18; nonzero, symmetric
$(19,3)$, $k=2$	identity (1.1); identity (3.2) for some j	fail ($\lvert Y\rvert$ relative 0.38; residual 0.99)
$(27,3)$, complex $k=10$	$B\bar{\not{n}}_i B^{-1} = \not{n}_i$; Majorana spinors in the corner	yes; none (corner of dimension 32)
$(11,3)$, $k=2$	$\dim K_+$, D'; identities (1.1), (3.2)	16, 10; hold, $j = -0.5000$
$(19,11)$, $k=10$	$\dim K_+$, D'; identities (1.1), (3.2)	16, 10; hold, $j = -0.5000$

Table D.1. Checks of Proposition 7.1 in explicit gamma matrices. Relative errors of the identities that hold are below 3×10^{-14}.

Appendix E. Sufficiency and other contractions in explicit gamma matrices

The program `contraction_check.py` in the repository checks Theorems 5.12 and 7.3 in the gamma matrices of Appendix B. For a real frame of rank k and a fixed multivector u it takes two random gauginos λ_1, λ_2 in the occupied corner, forms the operator $\epsilon \mapsto \sum_\mu \eta_{\mu\mu}\big(\gamma^\mu\lambda_2\, B(\epsilon, \gamma^\mu\lambda_1) + \gamma^\mu\lambda_1\, B(\epsilon, \gamma^\mu\lambda_2)\big)$ on $\widehat{K_{\chi'}}$, which is the polarisation of the left-hand side of (7.7), and fits it by least squares with the operators $\gamma(e_a \wedge u)$, $a = 1, \dots, D$. A relative residual at machine precision means that a source $J \wedge u$ absorbs the cubic terms for this pair of gauginos; a residual of order one means that no source of this form does. The contractions are $u = 1$, products of spacelike basis vectors, and $u = n_1 \wedge \cdots \wedge n_k \wedge w$ with w a product of residual basis vectors; indices refer to the orthonormal basis $e_0, \dots, e_{D-1}$ in which $e_0, \dots, e_{t-1}$ are timelike, and the frame is $n_i = (e_{t+i-1} + e_{i-1})/\sqrt{2}$, $m_i = (e_{t+i-1} - e_{i-1})/\sqrt{2}$. For Theorem 5.12 the program also evaluates the residual bilinears $\delta A_{\mu'}$ and $\xi^{\mu'}$ of (N3) and (N4) for random parameters and checks that ξ is symmetric. All 32 checks pass; Table E.1 lists them. They confirm the written proofs and do not replace them.

Appendix F. Checks for residual dimension two and several gauginos

The program `residual_two_check.py` checks Theorem 5.14 and Proposition 7.5 in the gamma matrices of Appendix B, with the real frame $n_i = (e_{t+i-1} + e_{i-1})/\sqrt{2}$, $m_i = (e_{t+i-1} - e_{i-1})/\sqrt{2}$ of Appendix E. For Nishino's transformations (5.16) with random field strengths obeying $F(n_i, \cdot) = 0$ it checks that $\delta\lambda$ stays in K, that the residual components of δA and ξ are nonzero and that ξ is symmetric, and that the commutator of two supersymmetries with Grassmann parameters is the translation by ξ: on A for every vector in V' and in the span of the n_i, and on λ for random derivatives $D_\mu\lambda$ in K with $D_{n_i}\lambda = 0$, both on shell and, for $D' = 2$, off shell. The commutators are written as symmetric parts of explicit matrices,

signature, rank	contraction u	D'	result (relative residual)
$(9,1)$, $(5,5)$, $k=0$	1	10	absorbed (2×10^{-16})
$(9,1)$, $k=0$	e_9	10	not absorbed (1.00)
$(7,1)$, $k=0$	$e_1\wedge\cdots\wedge e_7$; 1	8	absorbed (3×10^{-16}); not absorbed (1.00)
$(11,1)$, $k=0$	1; e_1; $e_1\wedge\cdots\wedge e_9$	12	not absorbed (1.00; 0.93; 0.98)
$(11,3)$, $k=0$	$e_3\wedge\cdots\wedge e_{12}$	14	not absorbed (1.00)
$(9,7)$, $k=0$	e_7	16	not absorbed (1.00)
$(11,3)$, $k=2$	$n_1\wedge n_2$ (Sezgin); $n_1\wedge n_2\wedge e_{13}$	10	absorbed (1×10^{-15}); not absorbed (1.00)
$(11,3)$, $k=1$	n_1; $n_1\wedge e_{12}$	12	not absorbed (1.00; 0.91)
$(19,3)$, $k=2$	$n_1\wedge n_2$; $n_1\wedge n_2\wedge e_{21}$	18	not absorbed (0.99; 1.00)
$(19,3)$, $k=3$	$n_1\wedge n_2\wedge n_3\wedge e_{12}\wedge\cdots\wedge e_{21}$	16	not absorbed (1.00)
$(19,3)$, $k=1$	$n_1\wedge e_{14}\wedge\cdots\wedge e_{21}$	20	not absorbed (1.00)
$(9,1)$, $(5,5)$, $(1,9)$, $k=0$	$\delta A_{\mu'}$, $\xi^{\mu'}$; $u=1$	10	nonzero, symmetric; absorbed
$(7,7)$, $k=2$; $(3,11)$, $k=2$	$\delta A_{\mu'}$, $\xi^{\mu'}$; $u=n_1\wedge n_2$	10	nonzero, symmetric; absorbed
$(2,10)$, $k=1$	$\delta A_{\mu'}$, $\xi^{\mu'}$; $u=n_1$	10	nonzero, symmetric; absorbed
$(9,9)$, $k=4$	$\delta A_{\mu'}$, $\xi^{\mu'}$; $u=n_1\wedge\cdots\wedge n_4$	10	nonzero, symmetric; absorbed

TABLE E.1. Checks of Theorems 7.3 (upper part) and 5.12 (lower part) in explicit gamma matrices. In the lower part the asymmetry of ξ is below 4×10^{-14} and the residuals of absorption below 3×10^{-15}.

and the coefficient of the translation, which is fixed by the closure on A, is the same on λ. It then evaluates the cubic term $\sum_\mu \eta_{\mu\mu}\big(\gamma^\mu\lambda_2\,B(\epsilon,\gamma^\mu\lambda_1)+\gamma^\mu\lambda_1\,B(\epsilon,\gamma^\mu\lambda_2)\big)$ for every ϵ in the Weyl module of the parameter, and the cross term $\sum_\mu \eta_{\mu\mu}\gamma^\mu\sigma\,B(\epsilon,\gamma^\mu y)$ of two independent gauginos $\sigma,y\in K$. Nishino's theories in $(9,1)$ with $k=0$ and $(10,2)$ with $k=1$ serve as controls. All 50 checks pass; Table F.1 lists them. They confirm the written proofs and do not replace them.

signature, rank	D'	check	result
$(1,1)$, $k=0$; $(2,2)$, $k=1$	2	$\delta\lambda\in K$; (N3), (N4)	holds; nonzero, symmetric
$(3,3)$, $k=2$; $(4,4)$, $k=3$	2	closure on A	translation by ξ (residual $\le 3\times10^{-16}$)
		closure on λ, on and off shell	translation by ξ (residual $\le 3\times10^{-16}$)
		cubic term; cross term	both vanish ($\le 4\times10^{-16}$)
$(9,1)$, $k=0$; $(10,2)$, $k=1$	10	$\delta\lambda\in K$; (N3), (N4); closure on A and λ	hold (residual $\le 3\times10^{-16}$)
		cubic term; cross term	both nonzero
$(3,1)$, $(2,2)$, $k=0$; $(3,3)$, $k=1$	4	cross term	nonzero
$(5,1)$, $(3,3)$, $k=0$; $(4,4)$, $k=1$	6	cross term	nonzero
$(7,1)$, $k=0$; $(5,5)$, $k=1$; $(6,6)$, $k=2$	8	cross term	nonzero
$(9,1)$, $k=0$	10	cross term	nonzero

TABLE F.1. Checks of Theorem 5.14 (upper part, with Nishino's theories as controls) and of Proposition 7.5 (lower part) in explicit gamma matrices. A nonzero cross term is of the size of the product of the two gauginos, at least 0.9 times it in every random sample.

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